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Optimal scheduling of smart home energy systems: A user-friendly and adaptive home intelligent agent with self-learning capability

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ABSTRACT

This paper proposed a user-friendly and adaptive home intelligent agent with self-learning capability for optimal scheduling of smart home energy systems. The intelligent agent autonomously identifies model parameters based on system operation data, eliminating the need for manual input and making it more user-friendly and practical to implement. It can also self-learn the latest energy consumption information from an updated dataset and adaptively adjust model parameters to accommodate changing conditions. Utilizing these determined models as input, the intelligent agent performs day-ahead optimal scheduling using the proposed many-objective integer nonlinear optimization model and automatically controls system operation. Experimental studies were conducted on a laboratory-based smart home energy system to verify the effectiveness of the developed intelligent agent in different scenarios. The results consistently demonstrate Mean Absolute Percentage Errors below -12.7 % across all three scenarios, indicating the accuracy of the intelligent agent. Furthermore, the optimal scheduling significantly enhances system performances. After optimization, daily operational costs, peak-valley differences, and CO_2 emissions were reduced by 34.1 % to 81.6 %, 29.2 % to 36.7 %, and 19.6 % to 43.2 %, respectively. Moreover, the PV generation self-consumption rate and self-sufficiency rate improved by 29.6 % to 38.0 % and 40.5 % to 49.4 %, respectively. The proposed intelligent agent provides invaluable guidance for optimal dispatch of smart home energy systems in real-world settings.

Introduction

Background

Renewable energy power generation technologies have been widely utilized in recently years, driven by the goal of carbon neutrality. As reported by the Nation Energy Administration, China's total installed capacity of photovoltaic (PV) power generation reached 520 million kW, as of October 2023 [1]. Although the large-scale applications of PV generation reduce carbon emissions, the intermittency and fluctuation of PV generation pose a great challenge to the stability of utility grids as the proportion of PV generation increases. Leveraging residential load flexibility can mitigate the impacts of PV generation on utility grids [2], associated with reducing the required capacity of battery storages [3] and enhancing system resilience [4]. For example, changing the temperature setpoints of thermostatically controlled loads (TCLs) or adjusting the operation time of postponed loads can reshape household electricity consumption patterns and profiles to provide grid services for utility grids [5,6].

To effectively manage these flexible loads, the deployment of a Home Energy Management System (HEMS) is imperative. A HEMS can be defined as a technological platform that enables to monitor, record, schedule, and control household flexible energy resources [7]. In a HEMS, all the models, such as prediction models, optimal models, and optimization algorithms, are programmed and packaged. Utilizing a HEMS can yield significant reductions in electricity bills and peak-valley differences of household power consumption profiles, along with improvements in PV generation self-consumption and self-sufficient rates [8]. Nevertheless, the development of HEMSs is a multifaceted challenge, involving the integration of models for smart home energy systems, occupant behaviors, optimization problem formulation, and solution methods [9]. Considerable efforts have been invested in addressing these complex issues to enhance the overall performances of

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Nomenclature

a	model parameters of photovoltaic power generation models
F	objective functions of optimization models
FiT	feed-in tariffs of distributed photovoltaic power generation
N	number of total time steps
P	power
t	time slot
X	decision variables of optimization models
Greek symbols	
ϵ_{pv}	unit capital cost of photovoltaic power arrays
$\epsilon_{g,im}$	time-of-use tariffs
λ	weights
μ	static carbon emission coefficient
ξ	dissatisfaction caused by adjusting flexible load operation
AC	air conditioner
EWH	electric resistance water heater
HEMS	home energy management system
MAE	mean absolute error
MAPE	mean absolute percentage error
PV	photovoltaic
RC	resistance capacitance
SCR	self-consumption rate
SSR	self-sufficiency rate
TCL	thermostatically controlled load

HEMSs.

Literature review

When deploying a HEMS in a real-world environment, the initial step involves establishing smart home energy system models, such as PV models and flexible load models (e.g. TCLs, postponed loads), etc., and developing occupant behavior models for flexible domestic appliances usage. Table 1 provides a summary of previous research on these models. As evident from it, existing studies often employed a simple efficiency model to represent PV generation [10,11,20,12–19], with some variations incorporating temperature corrections [21–24], the Sandia PV array performance model [25], the One-Diode Model [26], and the Characteristic Point Model [27]. In the case of TCLs, researchers widely utilized Resistance-Capacitance (RC) models [11,13,29,30, 14–16,18,21,24,25,28], and Physically-Based Models [10,20,23, 31–35]. Postponed loads, on the other hand, were often assumed to maintain constant operating power during operation [11,13,27,29,31, 32,34,36–40,14,41,15–18,20,21,24]. As for occupant behavior models, pre-defined schedules were typically used in previous studies [10,11,24, 25,27–34,13,36–40,14–18,21,23], as summarized in Table 1. Generally, system models and occupant behavior models usually comprise several parameters. For instance, in RC models for TCLs, parameters cover equivalent thermal resistance, equivalent thermal capacitance, cooling/heating power, energy conversion efficiency, dead-band, etc. Occupant behavior models include parameters for occupant thermal comfort, convenience, etc. These model parameters need to be defined within HEMSs in advance when deploying HEMSs in practice. In traditional HEMSs, these parameters were predetermined and manually input by engineers or end-users, which resulted in two drawbacks. On the one hand, the cumbersome parameter input is not only inefficient but also significantly diminishes user experience, making it unfriendly to end-users. On the other hand, it is challenging to identify these model parameters in a real-world environment, especially at scale, as the parameters vary from one household to another and are impacted by several factors, such as occupant age, gender, occupation, etc. Inaccurate model parameters would sacrifice the accuracy of models, thereby leading to the infeasible optimal operation schemes. These factors severely impede the practical engineering applications and widespread adoption of HEMSs. It is urgent to develop easy-to-deploy models for home energy systems and occupant behaviors, associated with corresponding methods that enable the automatic identification of model parameters.

In addition to the aforementioned issues regarding parameters configuration of smart home energy system models and occupant behaviors models for HEMSs, the adaptability and self-learning capabilities of HEMSs have also received considerable attention. In fact, some parameters of home energy system models and occupant behavior models are not fixed in the real-world environment but may vary with the changing external conditions. For example, PV generation efficiency tends to decrease over time due to dust deposition on the surface of PV panels [42]; the heating efficiency of electric resistance water heaters may reduce owing to scale buildup [43]; the operating power of flexible loads also fluctuates with the supply voltage [44]. Additionally, occupant behaviors of flexible household appliance usage vary with time, including working days, non-working days, seasons, etc. [45]. They also change in response to external factors like financial situations and the number of family members [46]. These facts highlight the high requirements for the adaptability and self-learning capabilities of HEMSs in engineering applications. HEMSs need to have the ability to self-learn the latest knowledge and adaptively adjust the model parameters according to the changed external conditions. By this way, the accuracy of the models can be maintained at a high level, contributing to more precise and effective optimal operation schemes. However, existing HEMSs lack adaptability and self-learning capabilities, as the parameters of system models and occupant behavior models are pre-defined, as discussed earlier. It is necessary to enhance the adaptability and self-learning capabilities of HEMSs.

When it comes to the optimization models for the optimal scheduling of smart home energy systems, substantial efforts have been dedicated to this field. As depicted in Table 1, previous studies widely employed mixed-integer linear programming models to address scheduling challenges [10,11,32–34,36–40,12,14–16,18,21,24,30]. The optimal dispatch problem was also expressed as a mixed-integer nonlinear programming problem [25,28], a linear programming problem [23], and a quadratic programming problem [17,20]. Objective functions, as pivotal components of these optimization models, have garnered significant attention. Researchers predominantly concentrated on optimizing end-users' benefits, including costs [10,11,20–29,12,30–34, 36–40,13,41,14–19], user comfort, discomfort, satisfaction, or dissatisfaction [10,11,29,31,33,36,37,41,13–15,17,19,20,25,27], which were widely applied as optimization objectives. However, the benefits of other stakeholders, for example grid operators and policymakers, were seldom considered. Solely prioritizing end-users' interests could disregard potential benefits of other stakeholders, preventing the practical applications of HEMSs. To strike a balance, it is imperative to take into account the benefits of diverse stakeholders.

As for the research methods for HEMS studies, an examination of the summarized data in Table 1 reveals that the majority of work has primarily relied on simulations. These simulations covered a range of cases and scenarios, serving as a way to validate the robustness and effectiveness of proposed models and algorithms [10,11,20–29,12,30–34, 36–39,41,13–19]. Although the simulation results can prove the feasibility of the methods to some extent, how these methods will perform in practical applications is still unknown due to the complicated operation conditions. Only a few studies were performed based on experiments.

Table 1
A summary of previous work and the present study.

Refs	System models			Occupant behaviors	Optimal models	Optimal objectives	Manage-ment	Research means
	PV	TCLs	Shiftable loads					
Langer et al. [10]	Simple efficiency models	Physically-based models	/	Pre-defined schedules	MILP	• Costs • Comfort	HEMS	Simulations
Huy et al. [11]	Simple efficiency models	RC models	Constant power models	Pre-defined schedules	MILP	• Costs • Comfort • PAR	HEMS	Simulations
Gonçalves et al. [31]	/	Physically-based models	Constant power models	Pre-defined schedules	Not mentioned	• Costs • Dissatisfaction	HEMS	Simulations
Yousefi et al. [25]	SAPM models	RC models	/	Pre-defined schedules	MINLP	• Costs • Comfort	HEMS	Simulations
Correa-Florez et al. [28].	/	RC models	/	Pre-defined schedules	MINLP	Costs	HEMS	Simulations
Dinh et al. [12]	Simple efficiency models	/	/	/	MILP	Costs	HEMS	Simulations
Duman et al. [21]	Simple efficiency models with temperature correction	RC models	Constant power models	Pre-defined schedules	MILP	Costs	HEMS	Simulations
Wang et al. [13]	Simple efficiency models	RC models	Constant power models	Pre-defined schedules	Not mentioned	• Costs • Satisfaction	IoT-based HEMS	Simulations
Tostado-Véliz et al. [14]	Simple efficiency models	RC models	Constant power models	Pre-defined schedules	MILP	• Costs • Comfort	HEMS	Simulations
Tostado-Véliz et al. [15]	Simple efficiency models	RC models	Constant power models	Pre-defined schedules	MILP	• Costs • Comfort • PAR • Convenience	HEMS	Simulations
Zupančič et al. [22]	Simple efficiency models with temperature correction	/	/	/	Not mentioned	• Costs • SSR	HEMS	Simulations
Alıç et al. [29]	/	RC models	Constant power models	Pre-defined schedules	Not mentioned	• Costs • Discomfort	HEMS	Simulations
Al Essa et al. [23]	Simple efficiency models with temperature correction	Physically-based models	/	Pre-defined schedules	LP	Costs	HEMS	Simulations
Esmaeel et al. [32]	/	Physically-based models	Constant power models	Pre-defined schedules	MILP	Costs	HEMS	Simulations
Wu et al. [33]	Not mentioned	Physically-based models	/	Pre-defined schedules	MILP	• Costs • Discomfort	HEMS	Simulations
Gomes et al. [30]	Not mentioned	RC model	/	Pre-defined schedules	MILP	Costs	HEMS	Simulations
Tostado-Véliz et al. [16]	Simple efficiency models	RC model	Constant power models	Pre-defined schedules	MILP	• Costs • Net energy consumption	HEMS	Simulations
Javadi et al. [36]	Not mentioned	/	Constant power models	Pre-defined schedules	MILP	• Costs • Discomfort	HEMS	Simulations
Duman et al. [24]	Simple efficiency models with temperature correction	RC model	Constant power models	Pre-defined schedules	MILP	Costs	HEMS	Simulations
Amabile et al. [35]	/	Physically-based models	/	Pre-defined schedules	Not mentioned	SCR	HEMS	Simulations
Javadi et al. [37]	/	/	Constant power models	Pre-defined schedules	MILP	• Costs • Discomfort	HEMS	Simulations
Ben et al. [26]	One diode model	/	/	/	Not mentioned	• Costs • Smoothing power profile	HEMS	Simulations
Nezhad et al. [38]	/	Constant power models	Constant power models	Pre-defined schedules	MILP	Costs	HEMS	Simulations
Sahoo et al. [39]	Input PV power generation	/	Constant power models	Pre-defined schedules	MILP	Costs	HEMS	Simulations
Ajao et al. [34]	Input PV power generation	Physically-based models	Constant power models	Pre-defined schedules	MILP	• Costs • CO_2 emissions	HEMS	Simulations

(continued on next page)

Table 1 (*continued*)

Refs	System models			Occupant behaviors	Optimal models	Optimal objectives	Manage-ment	Research means
	PV	TCLs	Shiftable loads					
Elkazaz et al. [40]	Input PV power generation	/	Constant power models	Pre-defined schedules	MILP	Costs	HEMS	Experiments
Jin et al. [17]	Simple efficiency models	Gray models	Constant power models	Pre-defined schedules	QP	• Costs • Comfort • Convenience • CO_2 emissions	HEMS	Simulations
Tostado-Véliz et al. [18]	Simple efficiency models	RC models	Constant power models	Pre-defined schedules	MILP	Cost	HEMS	Simulations
Rocha et al. [27]	Characteristic point models	Constant power models	Constant power models	Pre-defined schedules	Not mentioned	• Costs • Comfort	HEMS	Simulations
Zheng et al. [19]	Simple efficiency models	/	Actual operating power	Determined by monitored data	Not mentioned	• Costs • Comfort	HEMS	Simulations
Killian et al. [20]	Simple efficiency models	Physically-based models	Constant power models	Occupancy modeling and prediction	QP	• Costs • Comfort • Efficiency	HEMS	Simulations
Liu et al. [41]	/	Constant power models	Constant power models	Determined by monitored data	Not mentioned	• Costs • Comfort	HEMS	Simulations
This work	Gray models	Gray models	Gray models	• Statistical analysis • Stochastic occupant behavior models	INLP	Considering the interests of diverse stakeholders	Intelligent agent	Experiments

Abbreviation in the table: HEMS: home energy management system; LP: linear programming; MILP: mixed integer linear programming; MINLP: mixed integer nonlinear programming; MIQP: mixed integer quadratic programming; PAR: peak-to-average ratio; QP: quadratic programming; RC: resistance-capacitance; SCR: self-consumption rate; SSR: self-sufficiency rate; SAPM: Sandia PV array performance.

For instance, Elkazaz et al. [40] developed a hierarchical two-stage energy management approach for smart home energy systems, and validated it by experiments. Nevertheless, a limitation is that flexible loads were simulated using electronic load emulators during the experiments. A two-level control scheme was proposed for a building energy system equipped with a heat pump and warm and cold water storage tanks to provide frequency reserves, which was verified by experiments on an actual energy system [47]. Moreover, Zhang et at. [48] conducted a long-term field test to evaluate the performances of a model predictive control controller in a real-world commercial building equipped with PV and batteries. However, these studies [47,48] were conducted for commercial building energy systems, and experiment research for the optimal scheduling of smart home energy systems is lacking. Despite extensive studies conducted in the field of HEMSs, many of them have been rooted in theoretical exploration, centered around simulation-based perspectives. Only a few experimental studies have been conducted, and the actual performances of HEMSs in the real-world environment remain unclear. Furthermore, the technical pathways for applying HEMSs in engineering practices are not yet clear.

Motivation and contributions

The preceding literature review demonstrates that great efforts have been devoted to HEMSs in previous studies. Nevertheless, several research gaps still exist.

- The configuration of parameters for HEMSs in a real-world environment presents significant challenges and can be excessively burdensome, resulting in a diminished user experience.
- Traditional HEMSs lack adaptability and self-learning capabilities. They cannot adaptively adjust the model parameters according to the changed external environment, leading to the inaccurate models, which further cause infeasible optimal operation schemes.
- The majority of day-ahead optimal scheduling optimization models focused solely on end-user benefits, overlooking the interests of grid operators and policymakers, thereby compromising potential stakeholder advantages.
- Previous research on HEMSs primarily remains at the theoretical stage, relying on simulation methods, with only a few experiment studies conducted. The actual performances of HEMSs in the real-world environment are still unclear, and the technical pathways for applying HEMSs in engineering practices are not yet clear.

The main aim of this study is to develop an easy-to-deploy technology for optimal scheduling smart home energy systems in a real-world environment and provide a standardized pathway for its practical implementation. To achieve this, a home intelligent agent, which outperforms HEMSs in autonomy, convenience, self-learning, adaptability, and personalized service [49,50], was developed in this paper based on our proposed theoretical framework for optimal dispatch of smart home energy systems[51]. A comprehensive comparison between this work and existing studies is presented in Table 1. Compared with previous research, this work offers two major innovations and contributions:

- A user-friendly and adaptive home intelligent agent was proposed for the optimal dispatch of smart home energy systems, featuring self-learning capabilities. The intelligent agent possesses the following characteristics: (a) It enables to automatically identify the model parameters of systems and occupant behaviors with the personalized setting for various households realized, avoiding the cumbersome parameter setting in the traditional HEMSs; (b) It is capable of adaptively adjusting the model parameters with time and external changes, thus the accuracy of the models can be always maintained; (c) The interests of diverse stakeholders, including end-users, grid operators, and policymakers, were considered in the proposed many-objective integer nonlinear programming models embedded in the intelligent agent, thereby achieving a balance among the benefits of different stakeholders.

- Experiments were designed and performed to verify the effectiveness of the developed intelligent agent using in a real smart home energy system in a laboratory-based environment. This approach addresses the limitations of simulation-based studies and enhances the credibility of our findings. Additionally, we provide a standardized pathway for deploying the intelligent agent in practical applications through the construction of the experimental platform and the testing process. This guidance facilitates the real-world implementation of the intelligent agent in practical engineering.

Methodology

This section begins with an introduction to a typical smart home energy system. Following that, we provide a comprehensive overview of intelligent agents. Finally, we delve into the specifics of our proposed intelligent agent designed for optimizing smart home energy systems.

A typical smart home energy system

A typical smart home energy system is presented in Fig. 1. PV arrays are equipped to meet user's electricity demands, including flexible loads and non-flexible loads. The flexible loads are TCLs (e.g. air conditioners (ACs), electric resistance water heaters (EWHs)), and postponed loads (e.g. washing machines, clothes dryers, and dishwashers). Non-flexible loads consist of household appliances that cannot be actively managed, such as televisions, kitchen hoods, electric cookers, and refrigerators, etc. When PV generation is enough, the excessive is injected into utility grids, while in case of a shortage, electricity is imported from utility grids. The operation of the system is managed by our proposed home intelligent agent, which serves as the central control unit for the smart home energy system. The intelligent agent automatically performs day-ahead optimal scheduling of the system at a fixed time each day, and controls the operation of the system on the following day according to the obtained operation schemes.

Intelligent agent

An intelligent agent, often simply referred as an "agent", is a concept in artificial intelligence (AI) and computer science. An intelligent agent is an autonomous entity that can perceive its environment, make decisions, and take actions to achieve specific goals or objectives. It is usually used to model and solve complex problems that require decision-making and adaptation to changing conditions. Some of the key characteristics of intelligent agents include [52]:

- Autonomy: Intelligent agents possess self-governance and the ability to make decisions independently, drawing upon their own programming, knowledge, and perception of the environment.
- Perception: Intelligent agents enable to gather information from their surrounding environment. This information can include data from the physical world, other agents, or input provided by users.
- Reasoning: Intelligent agents exhibit various forms of reasoning or problem-solving capabilities. They can process perceived information, apply logical thinking or algorithms, and make decisions based on their reasoning.
- Actuation: Intelligent agents have the capacity to take actions within their environment. These actions can encompass physical movements, such as manipulating a robot's actuators, or virtual actions, such as transmitting messages in a communication network.
- Goal-Directed Behavior: Intelligent agents typically possess defined goals or objectives that they strive to achieve. They take actions to optimize the likelihood of achieving these goals.
- Self-learning: Some intelligent agents can learn from their experiences or from data, continually improving their performances over time. Machine learning techniques are often employed to facilitate learning in these agents.

Intelligent agents have been deployed in various applications, including robotics, autonomous vehicles, virtual personal assistants, recommendation systems, automated trading, and more. In this study, intelligent agents were firstly introduced to develop a tool for the day-ahead optimal dispatch of smart home energy systems. The details of the proposed intelligent agent are introduced in the following sections.

Development of home intelligent agents

A user-friendly and adaptive home intelligent agent with self-learning capability was developed for optimal scheduling of smart home energy systems. In this section, the architecture of our proposed intelligent agent was introduced firstly. Then, the key issues, user data privacy and security, for applying the intelligent agent in engineering practices were discussed. Finally, the operation mechanism of the

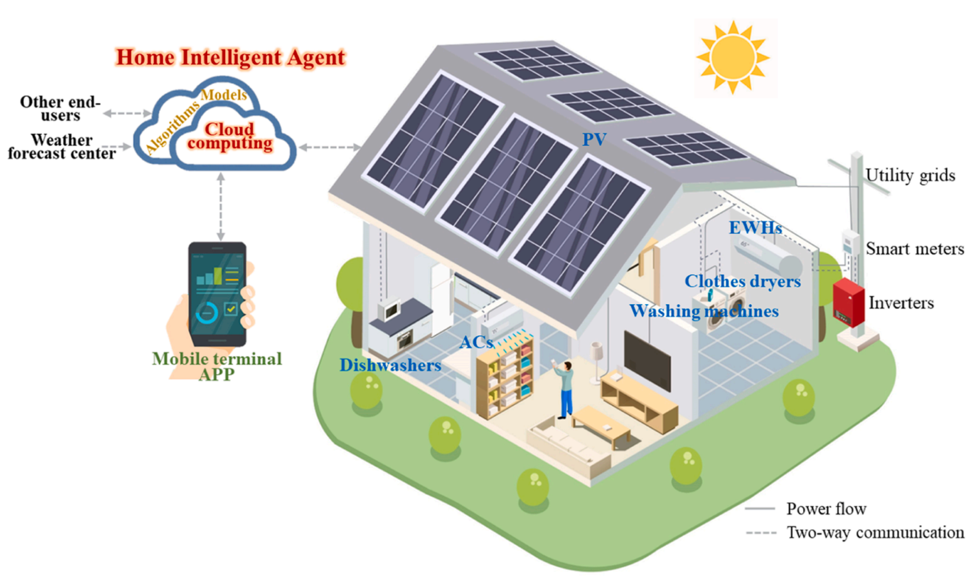


Fig. 1. Schematic diagram of a typical smart home energy system.

intelligent agent was described.

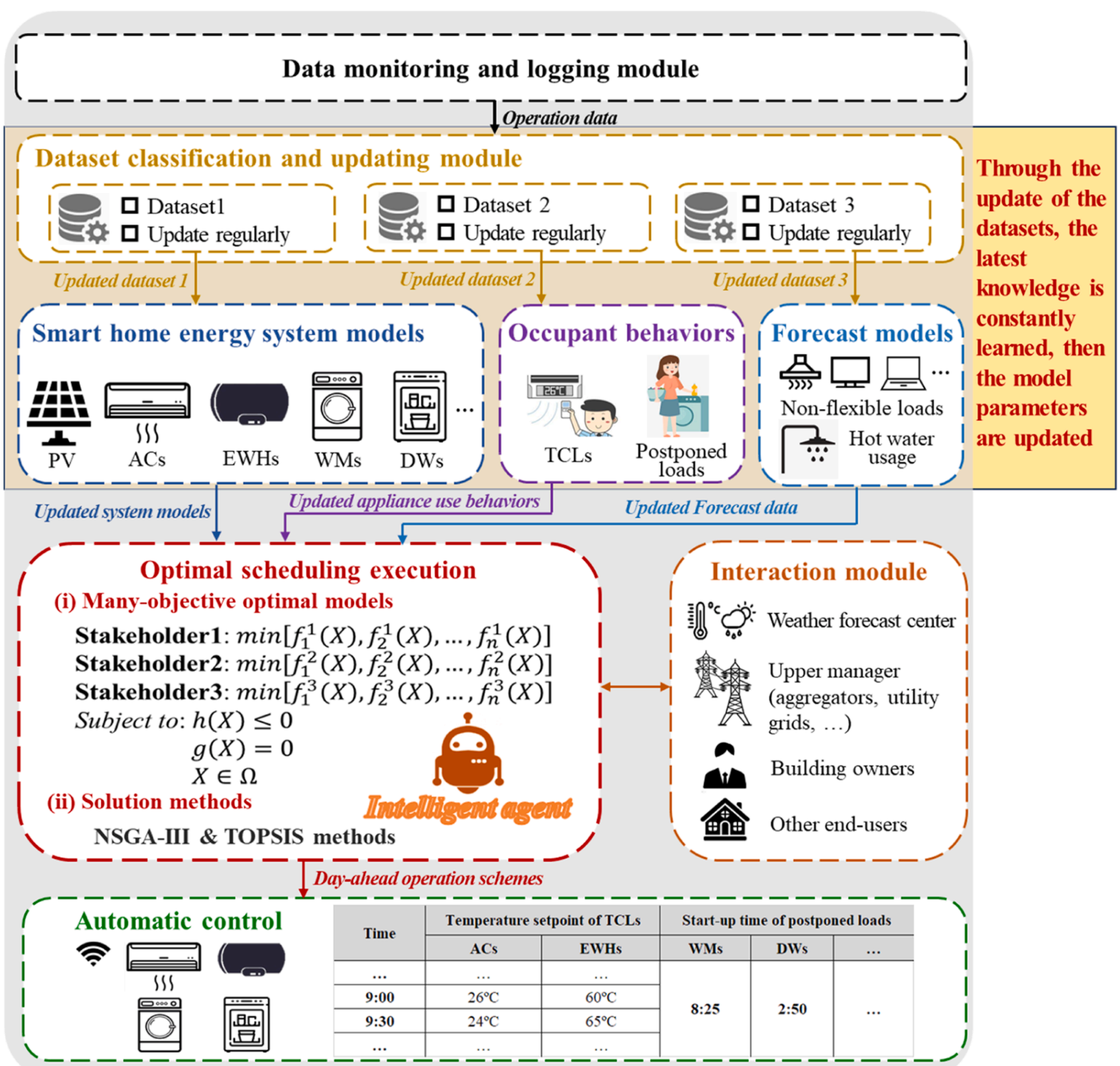
Architecture of home intelligent agent

The architecture of the developed home intelligent agent is presented in Fig. 2, which comprises eight modules: data monitoring and logging, dataset classification and updating, smart home energy system models, occupant behaviors, forecast models, interaction modules, optimal scheduling execution, and automatic control. The details are described as follows.

The data monitoring and logging module was designed to monitor and collect the operation data of the smart home energy system, providing a foundational dataset for subsequent model development. In the dataset classification and updating module, the monitored data was systematically categorized into three distinct datasets: Dataset 1, Dataset 2, and Dataset 3. These datasets were used to develop the smart home energy system models, occupant behavior models, and forecast models in the corresponding modules, respectively. The time window of each dataset was regularly updated, enabling the intelligent agent to self-learn the latest energy consumption information, then adaptively adjusting the model parameters to adapt to the changing conditions. Moreover, the interaction modules were designed to facilitates interactions between the intelligent agent and various entities, including the weather forecast center, aggregators or utility grids, building owners, flexible household appliances, and other households. Further details are presented in Appendix A. Utilizing the determined models and interactive data as inputs, the optimal scheduling execution module performs day-ahead optimal scheduling of the systems to obtain the optimal operation schemes. The automatic control module converts the acquired optimal operation schemes into execute commands and automatically controls the operation of flexible domestic appliances. The methods used to develop smart home energy system models, occupant behavior models, and optimal models will be introduced in Section 2.4.

User data privacy and security

For the application of the developed intelligent agent, user data privacy and security are key issues that ensure users' confidence in the ethical handling of their information, particularly due to the system's reliance on user data. To address these concerns, several measures were implemented to enhance data privacy and security within our research. Robust encryption techniques were employed to protect sensitive user data during transmission and storage. This included encryption key management, encryption strength, and adherence to industry-standard encryption protocols. Strict access control mechanisms were established to ensure that only authorized individuals have access to user data. Anonymization techniques, such as data masking, pseudonymization, and aggregation, were adopted to maintain the security and



Time	Temperature setpoint of TCLs		Start-up time of postponed loads		
	ACs	EWHs	WMs	DWs	…
…	…	…	8:25	2:50	…
9:00	26°C	60°C			
9:30	24°C	65°C			
…	…	…			

Fig. 2. Architecture of the user-friendly and adaptive intelligent agent with self-learning capability.

privacy of individual user information. Additionally, data protection regulations were followed with regular data protection impact assessments and robust data breach response protocols established. By incorporating these specific measures, user data privacy and security concerns were effectively addressed, promoting the practical application of the intelligent agent.

Operation mechanism of home intelligent agents

When deploying the intelligent agent in a household, the first step is to establish seamless communication channels between the agent and various components, such as smart meters, smart flexible household appliances, weather forecast centers, and users. Subsequently, the intelligent agent enters a data collection phase that typically spans a month. The collected data is used to identify model parameters, including those related to smart home energy systems and occupant behaviors. Importantly, these model parameters receive regular updates based on evolving datasets, namely Dataset 1, Dataset 2, and Dataset 3. Utilizing these determined or updated models as input, the intelligent agent performs the day-ahead optimal scheduling at a predefined time. The obtained operation schemes are then transmitted to the flexible household appliances, which autonomously control their operations accordingly on the following day. The pseudocode that describes the agent's autonomous functionality is presented below.

Algorithm 1. Algorithms for the agent's autonomous operation.

```
Initialize parameters, including day_0 and T_update (denoting the period at which
datasets are updated), etc.
get operation data of smart home energy systems
Classify the collected data into various datasets, namely Dataset1, Dataset2,
and Dataset3
output Dataset1, Dataset2, and Dataset3
for i in range (day_0, +∞, T_update):
   get and update Dataset1_i, Dataset2_i, and Dataset3_i
   Identify and update model parameters of system models, occupant behavior
models, etc. based on the updated Dataset1_i, Dataset2_i, and Dataset3_i,
respectively
   output the updated model parameters
   for day_j in range (i, i+ T_update, 1):
      get weather forecast data in day_j, such as ambient temperature and solar
radiation, electricity tariffs, etc.
      import the above updated model parameters into optimization models
      Conduct optimal scheduling calculations based on the optimization model
      output the optimal operation schemes in day_j
      Convert the acquired optimal operation schemes into execute commands
      Send these commands to the corresponding flexible domestic appliances
on the following day and automatically control their operation
      output display real-time operating status of domestic appliances and
system performances in APP of user terminals
   end for
end for
End
```

Problem formulation

The day-ahead optimal scheduling of smart home energy systems was formulated as a many-objective optimization problem. In this section, smart home energy system models and occupant behavior models that are used as constraints of optimization models were developed firstly. Then, a many-objective integer nonlinear optimization model was proposed to perform the optimal scheduling. All the models were programmed and packaged into the corresponding modules of the intelligent agent.

Smart home energy system models

Smart home energy system models were developed, including PV, ACs, EWHs, washing machines, dishwashers, etc. As mentioned above, existing HEMSs typically require users to manually input numerous model parameters, and obtaining these parameters can be challenging in a real-world environment. Therefore, the gray models of smart home energy systems were developed in this study. These models are easy to deploy and do not need users to manually input model parameters, which can be identified using historical operational data. As the dataset used for model development is updated, the model parameters are also updated accordingly, continuously adjusting the model to adapt to the changing external conditions. This ensures that the model maintains a high level of accuracy.

Taking the PV model development as an example, the detailed modeling process is as follows. A gray model for PV generation was developed based on a simplified efficiency model with temperature correction, as illustrated by Eq. (1). In this equation, $\boldsymbol{P_{pv}}$ denotes the PV generation power, while $\boldsymbol{Var}$ represents a variable linked to ambient temperature and solar radiation. The model is defined by two parameters, $\boldsymbol{a}_1$ and $\boldsymbol{a}_2$, which is significantly fewer than the parameters in the simplified efficiency model with temperature correction. These parameters can be accurately determined by linear regression methods based on Dataset 1, including PV generation power, ambient temperature, and solar radiation. Importantly, this approach does not require precise knowledge of PV array specifications, such as power generation efficiency, installed PV array dimensions, and inverter efficiency. This feature enhances the practical applicability of the model, especially when deployed across numerous households, as it eliminates the need for manual input of PV array details on a per-household basis. Another distinct advantage of this model lies in its adaptability. For example, as dust accumulates on PV panels, the power generation efficiency may decrease, subsequently affecting PV generation power. The developed model exhibits the capability to adaptively adjust its parameters to enhance model accuracy based on the updated Dataset 1, while the traditional simple efficiency model with temperature correction model does not possess this capability.

$$\boldsymbol{P_{pv}}(\boldsymbol{t}) = \boldsymbol{a}_1 \cdot \boldsymbol{Var}(\boldsymbol{t}) + \boldsymbol{a}_2 \tag{1}$$

Similarly, the gray models of TCLs and postponed loads were also developed. Based on Dataset 1, the model parameters can be identified by statistical analysis and linear regression methods, and they can be adaptively adjusted with the updated Dataset 1 to improve the model accuracy. The detailed models are presented in Appendix B.

Occupant behavior models

Occupant behaviors regarding the usage of flexible domestic appliances have a significant impact on the optimal scheduling of smart home energy systems. In this study, we considered two types of occupant behaviors: "static occupant behaviors" and "dynamic occupant behaviors".

Static occupant behaviors describe the habits of occupants regarding the usage of flexible household appliances. For example, they include habits related to the temperature setpoints of TCLs and the preferred timing for using postponed household appliances. The former reflects occupant preferences for thermal comfort with TCLs, while the latter indicates occupant preferences for convenient use of postponed loads. To ensure occupant thermal comfort and convenience, it is crucial that the day-ahead optimal scheduling aligns with these static occupant behaviors. The static occupant behaviors are defined by two sets of parameters: $\left[\boldsymbol{T}_{tcl}^{set,\ min}, \boldsymbol{T}_{tcl}^{set,\ max}\right]$ and $\left[\boldsymbol{t}_{shift}^{start,\ min}, \boldsymbol{t}_{shift}^{start,\ max}\right]$. Here, $\boldsymbol{T}_{tcl}^{set,\ min}$ and $\boldsymbol{T}_{tcl}^{set,\ max}$ represent the minimum and maximum temperature setpoints for TCLs, while $\boldsymbol{t}_{shift}^{start,\ min}$ and $\boldsymbol{t}_{shift}^{start,\ max}$ denote the earliest and latest switch-on time for postponed loads. These parameters are determined using statistical analysis methods based on the data from Dataset 2, they can be adaptively adjusted with the updated Dataset 2.

Dynamic occupant behaviors include various aspects of occupant interactions with flexible household appliances, including usage frequency, activation time, operation durations and temperature setpoints for TCLs, as well as usage frequency, operation modes, and activation time for postponed loads. These occupant behaviors exhibit variations across different days and evolve over time, which need to be predicted in advance when performing day-ahead optimal scheduling. Consequently, a stochastic occupant behavior model was developed in our previous study [53] to predict day-ahead dynamic occupant behaviors, using the Metropolis Hasting algorithm-based Markov chain Monte-Carlo method. In this study, the model was adopted to forecast the dynamic occupant behaviors. More details regarding the model can be found in Ref [53]. The stochastic occupant behavior model was trained using the historical operation data of TCLs and postponed loads, which are contained in Dataset 2, and it is capable of adaptively adjusting the model parameters to improve model accuracy according to the updated Dataset 2.

Optimization models and solution methods

A many-objective inter nonlinear optimal model was developed to formulate the problem of day-ahead optimal dispatch of smart home energy systems, considering the interests of various stakeholders. The details are described below.

The core decision variables of the optimization model are TCL's temperature setpoints T_{tcl}^{set}, and postponed load's switch-on moments t_{shift}^{start}, which are integer variables, as given by Eq. (2). The total number of the decision variables inherently depends upon the quantity of flexible loads incorporated within the system, increasing as the number of flexible loads increases. It also depends on occupant behaviors related to TCL usage, such as the switch-on moment t_{tcl}^{start} and switch-off moment t_{tcl}^{end}, as TCLs cannot be scheduled when they are turned off. A longer operation duration of TCLs would result in an increased number of the optimization variables T_{tcl}^{set}.

$$X = \left[T_{tcl}^{set}\left(t_{tcl}^{start}\right), \dots, T_{tcl}^{set}\left(t_{tcl}^{end}\right), t_{shift}^{start}\right] \tag{2}$$

In addition to the decision variables, objective functions are also a key component of optimization models. Considering the fact that optimal scheduling involves the benefits of different stakeholder, including end-users, utility grids, and policymakers, six objective functions were considered in the many-objective optimization model. The objective functions aim to minimize daily operation costs, users' dissatisfaction, peak-valley differences of household net power profiles, and daily carbon emissions, while maximizing the self-consumption rate (SCR) and self-sufficiency rate (SSR). The details are described below.

- Minimize the daily operation costs F_{cost}. F_{cost} consists of the costs of PV arrays C_{pv}, electricity bills from purchasing power from utility grids, and the earnings from selling excessive PV generation to utility grids, as expressed by Eq. (3).

$$F_{cost} = min\left[P_{pv}^{nom}\epsilon_{pv}\Delta t\mathrm{T} \Big/ L_{cyc} + \sum_{t=1}^{N}\left(\epsilon_{g,im}(t)P_{grid,im}(t) - \mathrm{FiT}\cdot P_{grid,ex}(t)\right)\right] \tag{3}$$

where P_{pv}^{nom} is PV's rated capacity; L_{cyc} is PV's life time; N is the number of timesteps; ϵ_{pv}, $\epsilon_{g,im}$, and FiT are the initial costs of PV, electricity tariffs, and feed-in tariffs, respectively; $P_{grid,im}$ and $P_{grid,ex}$ are the power purchased from utility grids and the power sold to utility grids, respectively.

Minimize user's dissatisfaction $F_{dissatisfaction}$. User's dissatisfaction derives from adjusting TCLs' temperature setpoints and shifting postponed loads' operation time, which can be expressed by Eq. (4), according to Ref. [54]. In the equation, ξ_{tcl} and ξ_{shift} are the dissatisfaction caused by TCLs and postponed loads, as formulated by Eqs. (5)-(7).

$$F_{dissatisfaction} = min\left(\lambda_{tcl}\xi_{tcl} + \lambda_{shift}\xi_{shift}\right)$$
$$tcl \in \{ACs, EWHs\}; shift \in \{WMs, clothes\ dryers, dishwashers\} \tag{4}$$

$$\xi_{tcl} = \frac{\sum_{t_{tcl}^{start}}^{t_{tcl}^{end}} \frac{e^{\varphi_{tcl}(t)}-1}{e-1}}{N_{tcl}} \tag{5}$$

$$\varphi_{tcl}(t) = \frac{\left|T_{tcl}^{set}(t) - T_{tcl}^{set,pre}\right|}{T_{tcl}^{set,max} - T_{tcl}^{set,min}} \tag{6}$$

$$\xi_{shift} = \frac{\left|t_{shift}^{start} - t_{shift}^{start,pre}\right|}{t_{shift}^{start,max} - t_{shift}^{start,min}} \tag{7}$$

where λ_{tcl} and λ_{shift} are weights; φ_{tcl} indicates the degree to which the optimized TCL temperature setpoints $T_{tcl}^{set}(t)$ deviates from occupant preferred value $T_{tcl}^{set,pre}$; $\left[T_{tcl}^{set,min}, T_{tcl}^{set,max}\right]$ denotes occupant thermal comfort with respect to TCLs; $t_{shift}^{start,pre}$ denotes occupants desired switch-on moment of postponed loads; $\left[\tau_{shift}^{start,min}, t_{shift}^{start,max}\right]$ is the flexibility window of postponed loads.

Minimize the peak-valley differences of net power profiles $F_{peak-valley}$. The peak-valley differences were calculated by Eq. (8), where P_{grid} is net power of households.

$$F_{peak-valley} = min\{max\{P_{grid}(t)\} - min\{P_{grid}(t)\}\} \tag{8}$$

Minimize the daily carbon emissions $F_{emission}$. $F_{emission}$ was formulated by Eq. (9), where μ is static carbon emission coefficient.

$$F_{emission} = min\left(\sum_{t=1}^{N}\mu P_{g,im}(t)\Delta t\right) \tag{9}$$

Maximize SCR of PV generation F_{SCR}. F_{SCR} was expressed by Eq. (10), where $P_{pv-load}$ denotes the power of PV generation that is directly consumed by households.

$$F_{SCR} = max\left(\frac{\sum_{t=1}^{N}P_{pv-load}(t)\Delta t}{\sum_{t=1}^{N}P_{pv}(t)\Delta t}\right) \tag{10}$$

Maximize SSR of PV generation

F_{SSR}. F_{SSR} was calculated by Eq. (11), where P_{load} is the sum of household flexible loads and non-flexible loads.

$$F_{SSR} = max\left(\frac{\sum_{t=1}^{N}P_{pv-load}(t)\Delta t}{\sum_{t=1}^{N}P_{load}(t)\Delta t}\right) \tag{11}$$

Constraints are another key component of optimization models. Regarding the many-objective optimization model developed in this study, the constraints are static occupant behavior constraints, dynamic occupant behavior constraints, and energy balance constraints.

The static occupant behavior constraints are presented by Eqs. (12) and (13). The optimized TCLs' temperature setpoints must keep within occupant thermal comfort range, while the optimized switch-on moment of postponed loads should stay within the flexibility window. Occupant thermal comfort range $\left[T_{tcl}^{set,\ min}, T_{tcl}^{set,\ max}\right]$ and the flexibility window $\left[\tau_{shift}^{start,min}, t_{shift}^{start,max}\right]$ were identified by the approaches described in Section 2.4.2.

$$T_{tcl}^{set,\ min} \leq T_{tcl}^{set}(t) \leq T_{tcl}^{set,\ max},\ t \in \left[t_{tcl}^{start}, t_{tcl}^{end}\right] \tag{12}$$

$$t_{shift}^{start,\ min} \leq t_{shift}^{start} \leq t_{shift}^{start,\ max} \tag{13}$$

Dynamic occupant behavior constraints were considered as the constraints of the optimization model. The optimal scheduling of flexible loads was restricted by occupant behaviors of flexible household appliance usage, such as the use frequency, switch-on moment, and operation duration of TCLs, and the operation mode and use frequency of postponed loads. These dynamic occupant behaviors were determined by the developed stochastic occupant behavior model in Section 2.4.2. The identified dynamic occupant behaviors were input into the many-objective optimization model to perform day-ahead optimal scheduling.

Energy balance constraints were shown by Eq. (14). The power flow of smart home energy system must obey the law of conservation of energy. In the equation, $P_{pv}(t)$, $P_{tcl}^{in}(t)$, and $P_{shift}^{in}(t)$ are the PV generation power, TCLs operating power, and postponed load operating power, which were computed by corresponding models established in Section 2.4.1; $P_{nonflex}(t)$ is the power of non-flexible loads, which was predicted by persistence forecast methods using Dataset 3.

$$P_{grid}(t) = P_{tcl}^{in}(t) + P_{shift}^{in}(t) + P_{nonflex}(t) - P_{pv}(t) \tag{14}$$

The established many-objective inter nonlinear optimal model was solved by the reference-point-based Non-dominated Sorting Genetic Algorithm (NSGA-III) [55] to obtain Pareto solution set, using Platypus library in Python environment. A decision-making approach, Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) [56] method, was employed to determine the final solution from the Pareto solution set.

Experiment study

Experiments were carried out to validate our proposed models and methods in Section 2 in a real environment using a real smart home energy system. The details about the experiments are described in the following subsections, including the smart home energy system used in the experiments and the designs of the experiments.

Smart home energy system used in the experiments

A smart home energy system was configured in Building Energy integrated Smart Technologies Lab (BEST-Lab) at Hunan university, China. The schematic diagram of the smart home energy system is presented in Fig. 3. The main equipment of the system is displayed in Fig. 4, as described below:

Distributed PV generation: 10 kW PV panels, which are connected to a GW5000-DT inverter, were installed on the roof of the building. Moreover, a PV simulator was also configured, allowing the reproduction of the PV power curve for any day in the past. This simulator is helpful for conducting comparative or repetitive experiments.

Flexible loads: Flexible loads are a room AC, an EWH, a washing and drying machine, and a dishwasher, which can bi-directionally interact with the control platform of the smart home energy system via protocols.

- Non-flexible loads: Non-flexible loads comprise a refrigerator, an air purifier, a heater, a fan, etc.
- Data collection: A weather station was installed to collect environmental parameters, including solar radiation, environment temperature, wind speed, etc.; A smart meter was configured to monitor power exchanged with utility grids, including power imported from utility grids and power fed into utility grids; Smart sockets were used to monitor the operating power of flexible loads; Temperature sensors were adopted to monitor indoor room temperature and the ambient temperature around the EWH.
- Control platform: A control platform was developed to manage the operation of the smart home energy system with a local server and an upper computer configured. The local server is used to store data, provide computing services, manage information release and data, and support software development for the control platform; The upper computer provides functions, such as monitoring and control, data acquisition and processing, real-time display, remote operation and management, and man-machine operation interface. All the models and methods proposed in Section 2 were coded in Python and imported into the control platform to implement day-ahead optimal scheduling. The obtained optimal operation schemes were sent to flexible loads for automatic control on the next day. The upper computer displayed the real-time operation of the system for the following day. The control platform acted as the intelligent agent.

Experimental design

An extensive series of experiments were conducted to validate the efficacy of the developed intelligent agent. Three scenarios were set up based on the analysis of the history generation data of the roof PV at the BEST-Lab. In Scenario 1, PV generation power curve is a perfect parabola without fluctuation; PV generation is fluctuated in Scenario 2,

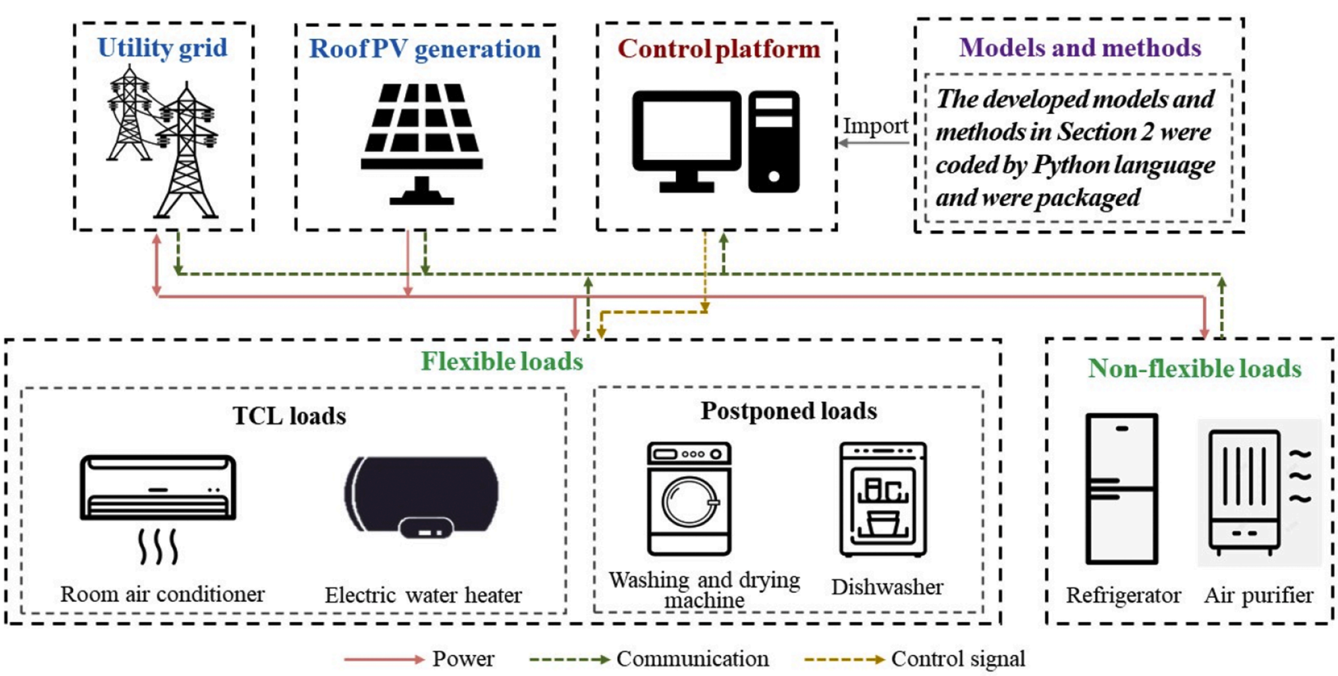


Fig. 3. Schematic diagram of the smart home energy system in the BEST-Lab at Hunan university.

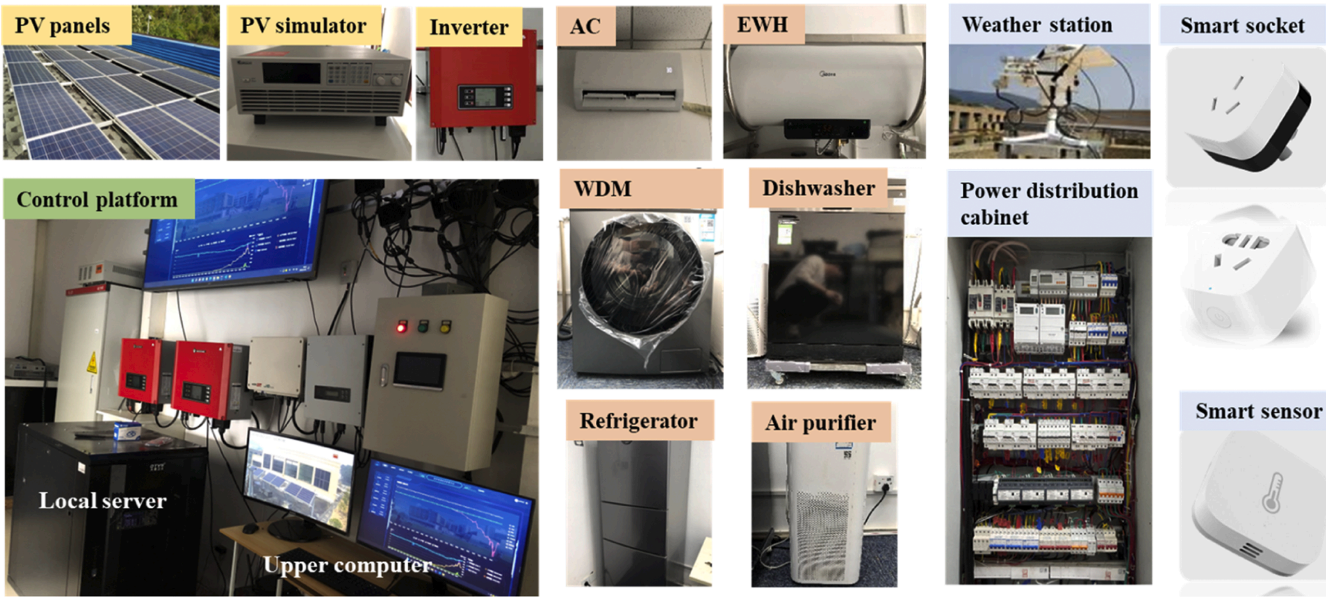


Fig. 4. Main equipment of the smart home energy system, used in the experiments, in the BEST-Lab at Hunan university.

while there is no PV generation in Scenario 3, as shown in Fig. 5. Each scenario was further divided into two cases: the baseline case and the optimized case. In the baseline case, the smart home energy system operated normally without being optimized, while the operation of the system was optimized using the developed models and methods in the optimized case. The specifics of these scenarios and cases are conveniently summarized in Table 2. During the experimental phase, cases within the same scenario were thoughtfully tested under similar weather conditions to mitigate the influence of daily weather variations. Moreover, the ambient temperature surrounding the EWHs was carefully controlled, maintaining consistent conditions via a room AC, to minimize environmental impacts on EWH performance. To ensure the robustness and reliability of the results, each case underwent experimentation three times, thus mitigating the influence of chance events and yielding more realistic outcomes. The experiments were conducted from July 20, 2023, to August 20, 2023.

Table 2
Scenarios and cases set up of the experiments.

Scenarios	Cases	Whether the system operation was optimized	PV generation
Scenario 1	Baseline case	No	PV generation power curve is a perfect parabola without fluctuation
	Optimized case	Yes	
Scenario 2	Baseline case	No	PV generation is fluctuated
	Optimized case	Yes	
Scenario 3	Baseline case	No	There is no PV generation
	Optimized case	Yes	

Experimental procedure

Before conducting the experiments, the model parameters of the smart home energy system models, occupant behavior models, and forecast models were identified. The historical operation data of the smart home energy system at the BEST-Lab was used to identify the system model parameters, including PV, a room AC, an EWH, a washing and dryer machine, and a dishwasher. The operation data of the system

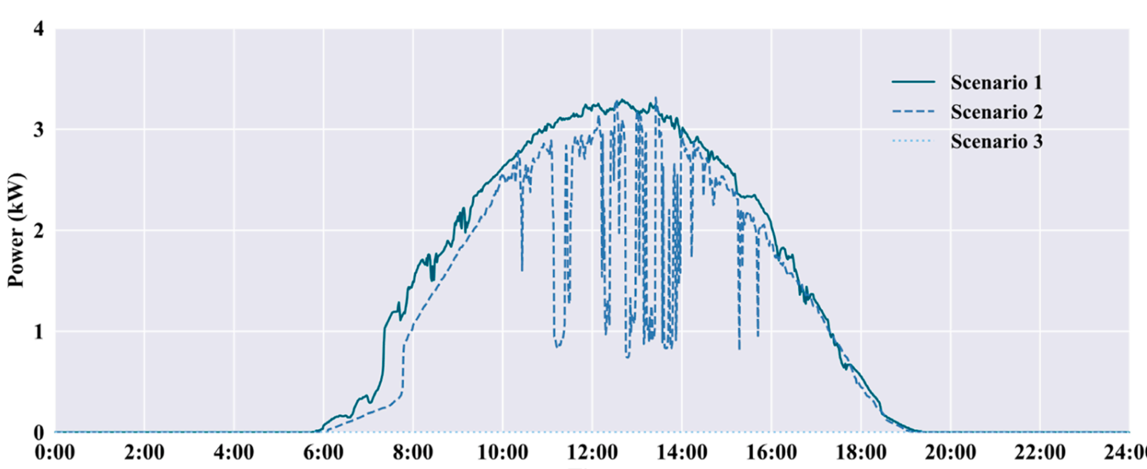


Fig. 5. PV generation power in various scenarios.

could not reflect the real-life energy use behaviors of a household, such as TCLs' temperature setting, postponed load usage, hot water consumptions, and non-flexible load power consumption. Therefore, the monitored data of an actual rural family in Changde (N 29°, E 112°), China, was employed to determine the model parameters of the occupant behavior models and the forecast models. The results of this model development are presented in Section 3.4.1.

Results

The results of the experiments are presented in this section. Firstly, the identified model parameters of the smart home energy system models and the occupant behaviors models are shown. Then, the experimental operation results of the smart home energy system under various scenarios are presented.

Model development results

According to the monitored operation data of the smart home energy system at the BEST-Lab, the model parameters of the system components were identified, as listed in Table 3. The indicators R-square (R^2) and Relative Root Mean Square Error (rRMSE) were adopted to evaluate the errors of the developed models. As illustrated in Table 3, the R^2 values for the PV models, AC models, and EWH models are 0.981, 0.996, and 0.997, respectively. The rRMSE values for the PV models, AC models, EWH models, washing and dryer machine models, and dishwasher models are 10.4 %10.4 %, 0.4 %, 0.3 %, 14.5 %, and 5.7 %, respectively. A comparison between the simulation results and the test results is presented in Fig. C.1 in Appendix C. It was found that the simulated operation curves agree well with the test ones. Hence, the developed models of the system are accurate.

Occupant behaviors of flexible domestic appliance use were also obtained. Regarding the static occupant behaviors, it was observed that the occupants are used to setting AC's temperature setpoints within [24, 29 °C] and EWH's temperature setpoints within [50, 75 °C], according to the statistical analysis results. Therefore, the determined $\left[T_{tcl}^{set,\ min}, T_{tcl}^{set,\ max}\right]$ of the AC is [24, 29 °C], while that of the EWH is [50 75 °C], as presented in Table 4. Similarly, the identified $\left[t_{shift}^{start,\ min}, t_{shift}^{start,\ max}\right]$ of the washing and dryer machine and the dishwasher are [05:00, 08:00] and [19:00, 06:00], respectively. Furthermore, the dynamic occupant behaviors were forecasted by the developed stochastic occupant behavior model, which was validated in our previous study [53], as listed in Tables 5 and 6, respectively. These determined parameters were input into the many-objective optimal model to conduct the day-ahead optimal scheduling.

Experimental results of Scenario 1

The experimental results of Scenario 1 were analyzed in this section, including the operation results of the baseline case and the optimized case. Moreover, a comparative analysis of the system performances under the two cases was carried out.

Table 3
Identified model parameters of PV and TCLs models.

Models	Identified model parameters	Errors	
		R^2	rRMSE
PV	$a_1 = 4.2166$, $a_2 = -84.6412$	0.981	10.4 %
AC	$b_1 = 0.989227$, $b_2 = 0.001074$, $b_3 = 0$, $b_4 = -0.000108$, $b_5 = 0.295523$, $P_{ac}^{on} = 910$, $P_{ac}^{off} = 19$	0.996	0.4 %
EWH	$c_1 = 0.999782$, $c_2 = 0.000218$, $c_3 = 0.00023$, $c_4 = -0.000292$, $c_5 = -0.000058$, $P_{ewh}^{on} = 2600$	0.997	0.3 %
Washing and dryer machine	Presented in Fig. C1(d) in Appendix C	/	14.5 %
Dishwasher	Presented in Fig. C1 (e) in Appendix C	/	5.7 %

Table 4
Static occupant behaviors of TCLs and postponed load usage.

Terms	$\left[T_{tcl}^{set,\ min}, T_{tcl}^{set,\ max}\right]$ of TCLs		$\left[t_{shift}^{start,\ min}, t_{shift}^{start,\ max}\right]$ of postponed loads	
	AC	EWH	Washing and dryer machines	Dishwashers
Behaviors	[24, 29 °C]	[50 75 °C]	[05:00, 08:00]	[19:00, 06:00]

Table 5
Dynamic occupant behaviors of TCL's usage on the next day.

TCLs	Dynamic occupant behaviors			Temperature setpoint at normal conditions
	Use frequency	Switch-on time	Operation duration	
ACs	1	09:00	900 mins	26 °C
EWHs	It was turned on throughout the day			65 °C

Table 6
Dynamic occupant behaviors of postponed load usage on the next day.

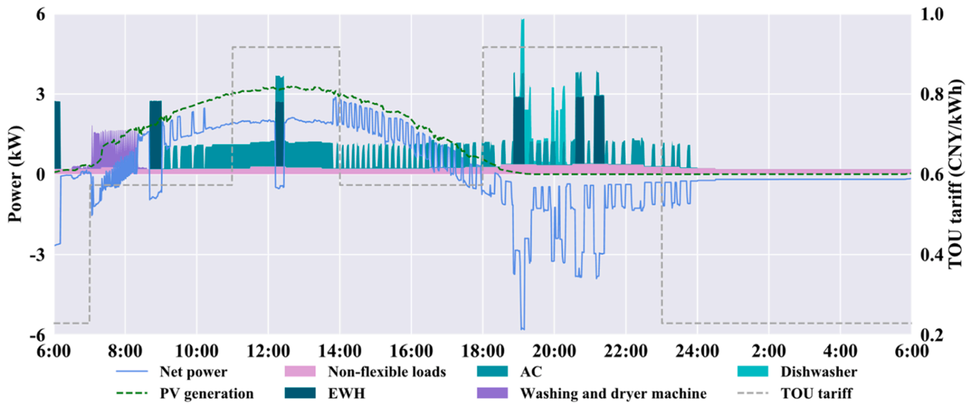
Postponed loads	Dynamic occupant behaviors		Turn-on moment at normal conditions
	Use frequency	Operation mode	
Washing and dryer machines	1	Normal	6:30
Dishwashers	1	Normal	19:00

Experiments of the baseline case

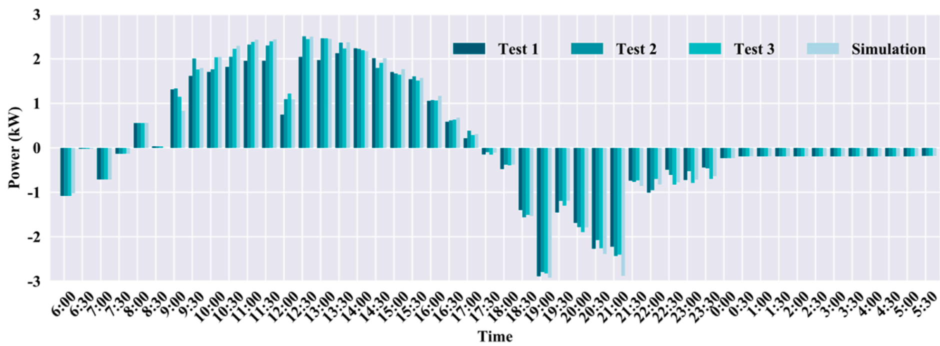
In the baseline case, the flexible household appliances operated normally with TCLs' temperature setpoints and postponed load's operation time set at normal values, as presented in Tables 5 and 6. The test of the baseline case was repeated three times. Taking one of the three tests as an example, the actual system operation throughout the day is illustrated in Fig. 6(a). As shown, the PV generation can satisfy the majority of the household's electricity demands during the daytime, while the household needs to purchase electricity from utility grids at night. Peak loads occur during the peak price periods, resulting from the usage of the AC, EWH, and dishwasher. Moreover, the net power of the system during the three tests is presented in Fig. 6(b) with a timestep of 30 mins, and the net power based on the simulation results is also displayed in this figure as well. As observed, the measured net power of the system during the experiments matches well with the simulated one. The Mean Absolute Errors (MAEs) between the three tests and the simulation are 140.6 W, 82.3 W, and 60.5 W, and the corresponding Mean Absolute Percentage Errors (MAPEs) are 2.9 %, 0.8 %, and 0.6 %. Therefore, our developed models are accurate.

Experiments of the optimized case

In addition to the experimental results of the baseline case, the test results of the optimized case in Scenario 1 were analyzed. The optimal operation schemes of the system, obtained by the intelligent agent, are illustrated in Fig. 7(a), including TCLs' temperature setpoints and the postponed loads' switch-on moments. As shown, TCLs' temperature setpoints were continuously changed but always remained within the occupant's preferred temperature setting range. Moreover, the switch-on moments of the postponed loads did not violate the user's usage



(a) Operation of the smart home energy system in the baseline case of Scenario 1 during one of the three tests.



(b) Tested and simulated household net power profiles of the baseline case of Scenario 1.

Fig. 6. Experimental results and simulation results of the baseline case in Scenario 1.

habits. Therefore, the obtained optimal operation schemes are consistent with the user's usage habits, making them feasible and applicable in practice.

The acquired operation schemes presented in Fig. 7(a) were implemented by the smart home energy system in the BEST-Lab at Hunan university to further verify their effectiveness in real-life world scenarios. Specifically, the control platform of the system sent the execute commands to the corresponding flexible domestic appliances to automatically control their operation. Taking one of the three tests as an example, the actual operation of the system throughout the day is demonstrated in Fig. 7(b). It was found that most TCLs and postponed loads were shifted to daytime operation as the PV generation was sufficient. The excess PV generation was injected into the utility grid with some revenue earned. It was also observed that the electricity consumption of the AC was minimized during high-prices periods at night for two reasons. Firstly, the temperature setpoints of the AC were adjusted to relatively high values owing to the high electricity price, while still remaining within the occupant's preferred temperature setting range, as illustrated in Fig. 7(a). Secondly, plenty of cooling was stored in the air-conditioning room during the daytime due to the thermal inertia of the room, which was released at night. As a result, the AC loads during high-price periods at night were curtailed. Moreover, the electricity demands at night were met by the utility grid because there was no PV generation.

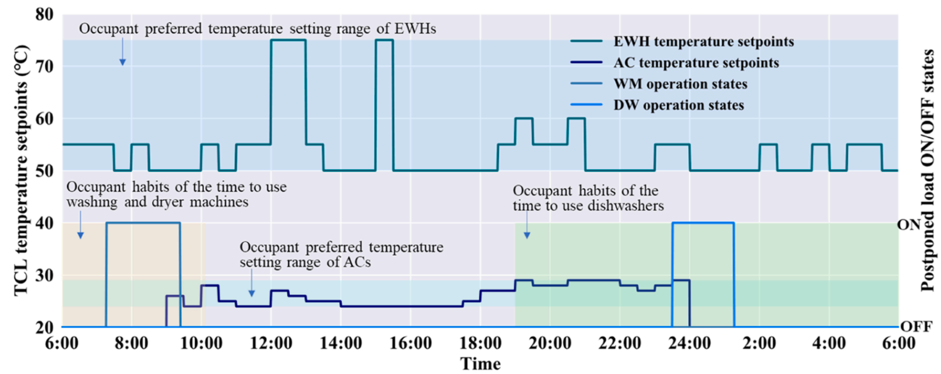
Additionally, the net power of the system during the three tests is presented in Fig. 7(c) with a timestep of 30 mins, and the net power based on the simulation results is displayed in this figure as well. As presented, the measured net power of the system during the experiments matches well with the simulated one. The MAEs between the three tests and the simulation are 103.7 W, 115.6 W, and 112.4 W, and the corresponding MAPEs are −6.6 %, −8.9 %, and −5.0 %. Therefore, our developed models are accurate.

Comparative analysis of the baseline case and the optimized case

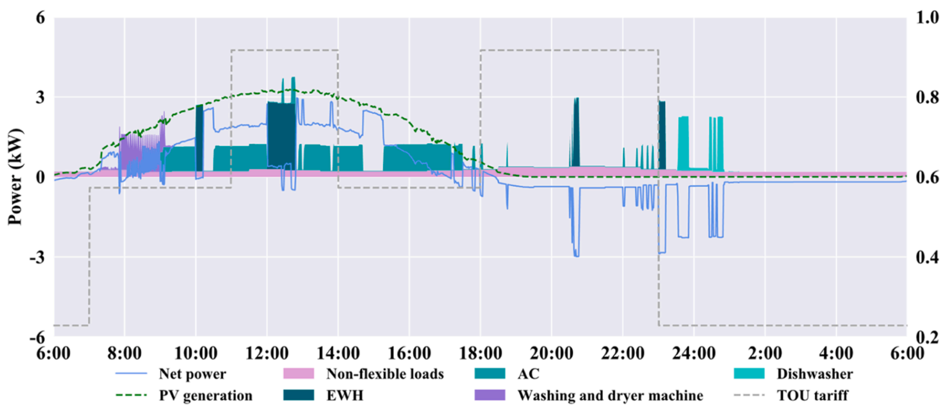
Based on the experimental results of the baseline case and the optimized case in Scenario 1, a comparative analysis of the system performances in terms of economy, environment, grid friendliness, and consumption of PV generation aspects was performed, as presented in Fig. 8. Compared to the baseline case, the daily operation costs, peak-valley differences, and CO_2 emissions in the optimized case were reduced by 81.6 %, 29.2 %, and 43.2 %, while the SCR and the SSR improved by 29.6 % and 40.5 %. After optimization, the system's performances were significantly enhanced, which can be explained by Fig. 9. In this figure, the net power of the system in various cases in Scenario 1 was displayed with a timestep of 30 mins. It was found that less PV generation was sold to utility grids in the optimized case, as indicated by the small areas enclosed by the net power curve, the x-axis, and the positive y-axis, filled in sky-blue in the figure, compared to the baseline case. This suggests that more PV generation was consumed by households. Moreover, electricity demands during peak-price periods in the optimized case were smaller than those in the baseline case, as denoted by the areas filled in red in the figure. These electricity loads filled in red were shifted to the daytime when PV generation is sufficient (filled in orange), enhancing the SCR, and shifted to the low-price periods (filled in pink), reducing electricity bills. Hence, the smart home energy system performs better in the optimized case.

Experimental results of Scenario 2 and Scenario 3

Similar conclusions can be drawn from the experimental results of Scenario 2 and Scenario 3. Errors between the experiments and the



(a) Optimal operation schemes of the system in Scenario 1.



(b) Operation of the smart home energy system in the optimized case of Scenario 1 during one of the three tests.

(c) Tested and simulated household net power profiles of the optimized case of Scenario 1.

Fig. 7. Experimental results and simulation results of the optimized case in Scenario 1.

simulations in the baseline and optimized case of Scenarios 2 and 3 are listed in Table 7. It was found that the MAEs of the two cases in Scenarios 2 and 3 are below 90 W, and the MAPE are below 14 %. Therefore, our developed models are accurate. Based on the experimental results, the system performances in the baseline and optimized case of Scenarios 2 and 3 were also calculated with the average value of the performance indicators obtained, which are presented in Table 8. In Scenario 2, the optimized case demonstrated a 51.3 % reduction in daily operational costs, a 30.9 % decrease in peak-valley differences, and a 41.0 % cut in CO_2 emissions compared to the baseline case, along with a 38.0 % increase in SCR and a 49.4 % increase in SSR. In Scenario 3, the optimized case presented reductions of 34.1 % in daily operational costs, 36.7 % in peak-valley differences, and 19.6 % in CO_2 emissions. The results reveal that the optimal scheduling can significantly enhance the performances of smart home energy systems.

Discussion

Specific contributions

A user-friendly and adaptive home intelligent agent with self-learning capability was proposed for optimal scheduling of smart home energy systems in this study. Compared to existing HEMSs [11,17,38], our innovative intelligent agent offers several compelling advantages. It enables to automatically identify model parameters without manually input, eliminating the need for cumbersome parameter setting in traditional HEMSs. Moreover, it can self-learn the latest energy consumption information and adaptively adjust the model parameters to accommodate changing conditions. Through these methods, the developed models of smart home energy systems and occupant behaviors demonstrate accuracy. Furthermore, the proposed many-objective optimization model takes into consideration the benefits of different

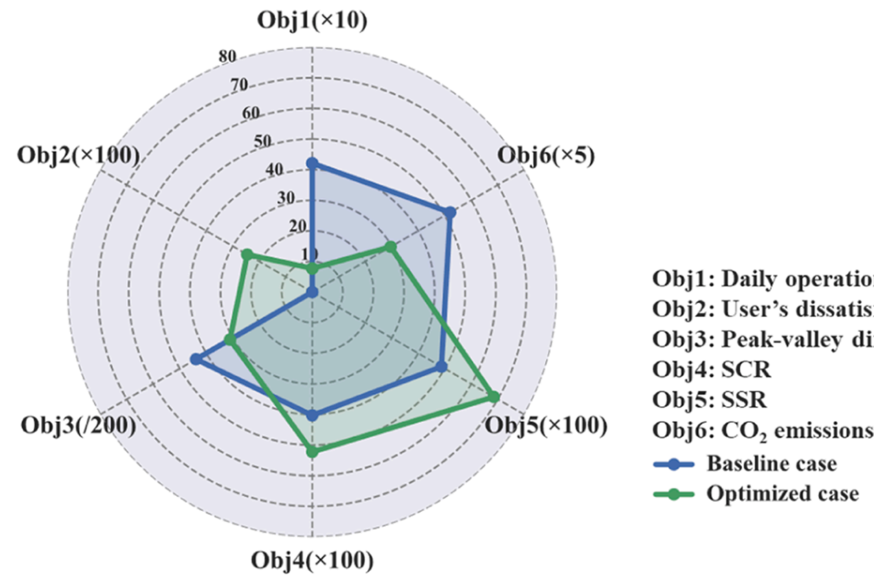


Fig. 8. Comparison of the system performances in various cases in Scenario 1.

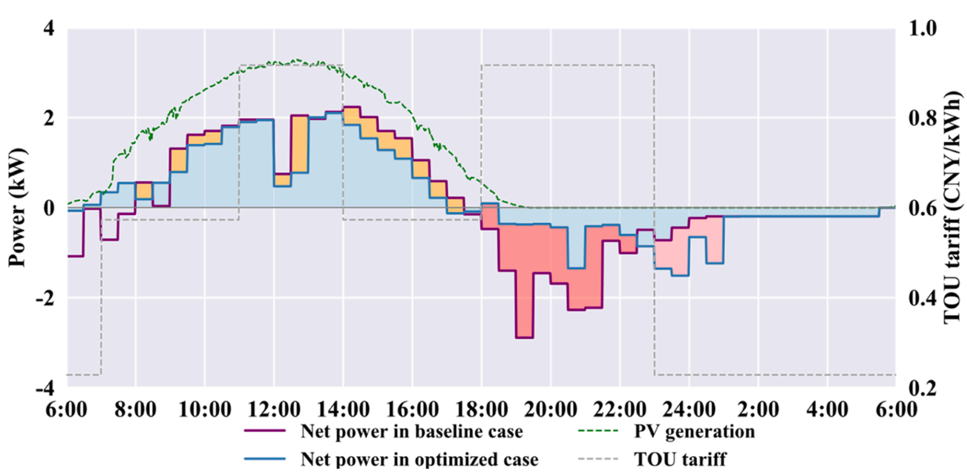


Fig. 9. Net power of the smart home energy system in various cases in Scenario 1.

Table 7
Errors between the experiments and the simulation in the baseline case and the optimized case in Scenario 2 and Scenario 3.

Scenarios	Cases	Test 1		Test 2		Test 3	
		MAE	MAPE	MAE	MAPE	MAE	MAPE
Scenario 2	Baseline case	77.4 W	−4.9 %	84.5 W	−1.7 %	54.9 W	−0.05 %
	Optimized case	74.8 W	−1.0 %	66.5 W	−12.3 %	66.5 W	−12.3 %
Scenario 3	Baseline case	81.0 W	−8.7 %	79.4 W	−9.3 %	57.5 W	−5.5 %
	Optimized case	85.8 W	−11.7 %	100.1 W	−13.7 %	94.3 W	−12.6 %

Table 8
Experimental system performances in Scenario 2 and Scenario 3.

Scenarios	Cases	Optimization objectives					
		SCR (%)	SSR (%)	Costs (CNY)	Dissatis-faction	Emissions (kg CO_2)	Peak-valley differences (W)
Scenario 2	Baseline case	43.02	42.32	6.47	0.00	11.47	8786.94
	Optimized case	59.38	63.21	3.15	0.29	6.77	6074.96
	Improvement	+38.0	+49.4	−51.3	/	−41.0	−30.9
Scenario 3	Baseline case	/	/	14.21	0.00	19.52	5631.77
	Optimized case	/	/	9.37	0.19	15.70	3563.92
	Improvement	/	/	−34.1	/	−19.6	−36.7

stakeholders, a factor seldom considered in existing HEMSs, achieving a tradeoff among the benefits of various stakeholders. For example, in Scenario 1, the optimization resulted in an 81.6 % reduction in daily operational costs, a 29.2 % decrease in peak-valley differences, and a 43.2 % reduction in CO_2 emissions, while the SCR and SSR improved by 29.6 % and 40.5 %, compared to the benchmark.

Experimental studies were conducted to verify the effectiveness of the developed intelligent agent and evaluate its performances in practice. In comparison to previous studies [10,11,20–29,12,30–34,36–39,41,13–19], which relied on simulation methods, our experimental studies overcome the limitations of simulation-based approaches and enhance the credibility of our findings. Furthermore, the actual performances of the intelligent agent in the real-world environment were clarified. After optimization, daily operation costs, peak-valley differences, and CO_2 emissions were reduced by 34.1 % to 81.6 %, 29.2 % to 36.7 %, and 19.6 % to 43.2 %, while PV generation self-consumption rate and self-sufficiency rate improved by 29.6 % to 38.0 % and 40.5 % to 49.4 %, respectively. Additionally, we provided a technical pathway for deploying the intelligent agent in practice, facilitating its real-world implementation in practical engineering.

Potential applications

A critical issue is the practical implementation of the proposed intelligent agent. To facilitate its deployment in real-world engineering applications, we have provided a standardized approach. The intelligent agent can be developed as a lightweight physical product or as a virtual one deployed in the cloud, as illustrated in Fig. 1. Once the intelligent agent is configured in a household, it operates according to the

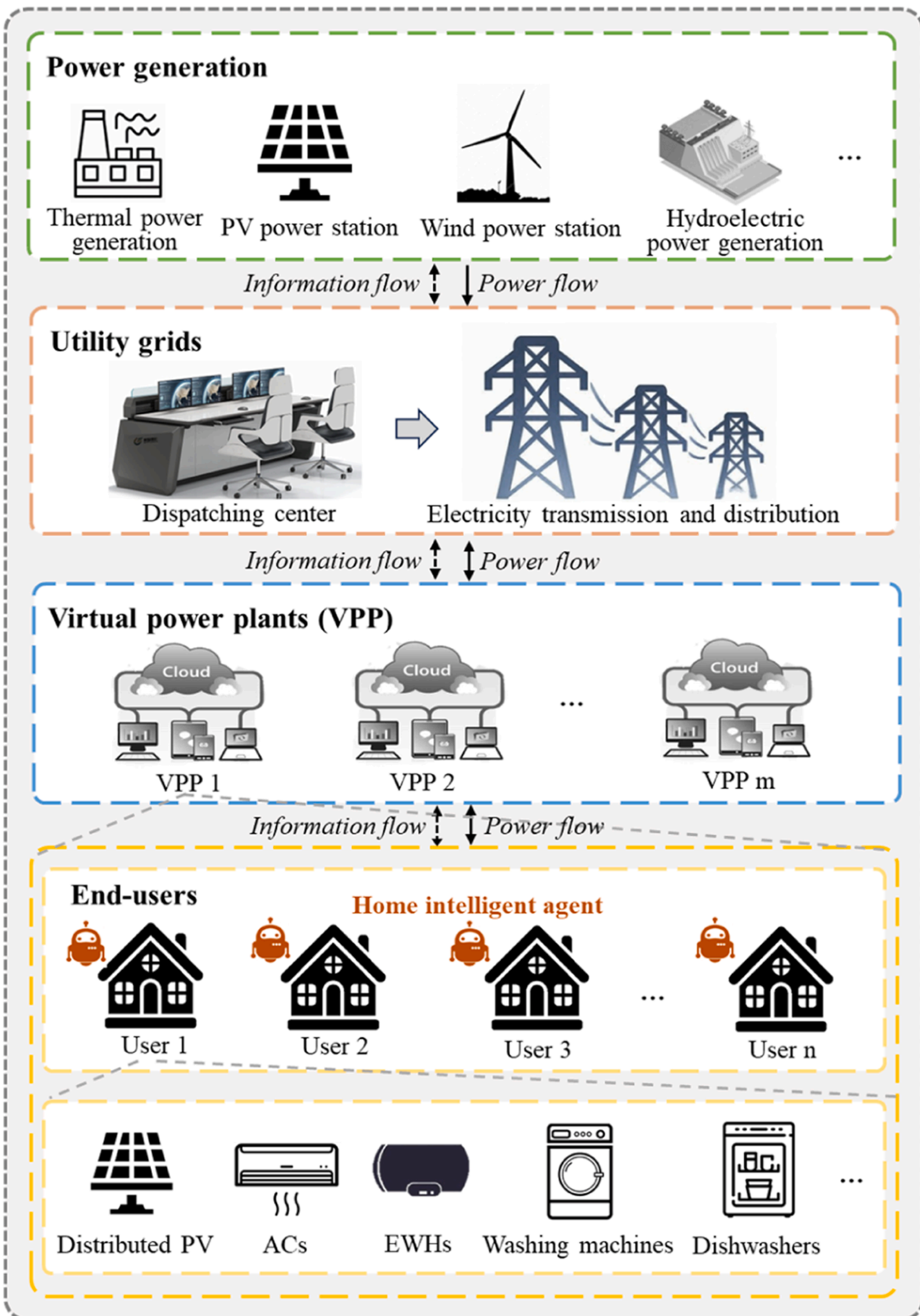


Fig. 10. A framework for the applications of the intelligent agent.

introduced operation mechanism. Importantly, although the experiments were performed in China based on a Chinese household, the framework is suitable for applications in other countries as well. The intelligent agent can also be applied to other countries. As the intelligent agent proliferates across numerous households, it has the potential to aggregate the distributed energy flexible resources from these households, forming a virtual power plant (VPP), as illustrated in Fig. 10. The VPP, coordinated by the intelligent agent in each household, can effectively manage these flexible resources to provide valuable services to utility grids. This is an area that we intend to investigate further in our future research.

Limitations and future work

This study was conducted with the assumption that the predictions, including occupant behaviors, non-flexible loads, and hot water consumptions, are accurate. However, uncertainties often arise during these forecasts, potentially resulting in suboptimal operation schemes and reduced system performances. We intend to address this challenge in our future research by coordinating day-ahead optimal scheduling with real-time operation optimization for smart home energy systems, while accounting for forecast uncertainties. This effort aims to further enhance the effectiveness of our proposed intelligent agent. Subsequently, we plan to develop this intelligent agent into a market-ready product and conduct extended real-world testing in multiple households over a span of, for example, two years. This initiative aligns with our goal of promoting practical engineering applications and contributing to the pursuit of carbon neutrality.

Conclusions

A home intelligent agent was proposed for optimal scheduling of smart home energy systems in a real-world environment in this paper. Experimental studies were conducted on a laboratory-based smart home energy system to validate the proposed intelligent agent in various scenarios. Additionally, a standardized deployment pathway for its practical engineering applications was outlined. The main conclusions are drawn below:

- The general gray models for distributed PV generation, thermostatically controlled loads, and postponed loads were developed based on their physical models. Model parameters of these proposed models can be identified using statistical analysis, multiple linear regression, and K-means clustering methods with history operation data. Experiments results demonstrate the accuracy of these developed models.
- A user-friendly and adaptive home intelligent agent was proposed for optimal scheduling of smart home energy systems. It adeptly identifies model parameters based on household operation data, and can adaptively update these model parameters over time to accommodate changing conditions. Experimental results prove the effectiveness of our developed intelligent agent with MAPEs between the experiments and simulations below −12.7 % across all three scenarios.
- Day-ahead optimal scheduling significantly improves the performances of smart home energy systems in terms of economy, environment, grid friendliness, and PV generation consumption aspects in real-world environments. After optimization, peak loads were curtailed or shifted to daytime periods coinciding with PV generation or low-price periods, thereby enhancing system performances. Experiment results in diverse scenarios reveal that optimal scheduling contributed to reductions 34.1 % to 81.6 % in daily operation costs.

CRediT authorship contribution statement

Zhengyi Luo: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Jinqing Peng:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization. **Xuefen Zhang:** Writing – review & editing, Resources, Project administration, Funding acquisition, Formal analysis. **Haihao Jiang:** Project administration, Funding acquisition. **Rongxin Yin:** Writing – review & editing, Formal analysis. **Yutong Tan:** Writing – review & editing, Formal analysis. **Mengxin Lv:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A

The interaction module was designed to facilitate interactions between the intelligent agent and various entities, including the weather forecast center, aggregators or utility grids, building owners, flexible household appliances, and other households. The details are below.

- Interaction with weather forecast center. The intelligent agent communicates with the local weather forecast center by an Application Program Interface to obtain the forecasted meteorological data, such as ambient temperature, solar radiation, etc.
- Interaction with utility grids. The intelligent agent interacts with utility grids to receive the electricity tariffs, exporting the excessive PV generation to utility grids, or importing electricity from utility grids.
- Interaction with building owners. The communication between the intelligent agent and building owners was realized by the APP, which is installed on the user's mobile terminal. For example, users can set or modify static occupant behavior parameters, including $\left[\boldsymbol{T}_{tcl}^{set,\ min}, \boldsymbol{T}_{tcl}^{set,\ max}\right]$ and $\left[\boldsymbol{t}_{shift}^{start,\ min}, \boldsymbol{t}_{shift}^{start,\ max}\right]$ through the APP (if the user does not set the above parameters, statistical analysis results are used in the optimization scheduling process). Users can view the operating status of domestic appliances at any time through the APP, such as AC operation mode, temperature setpoint, etc., and can also view system performances in economy, environment, grid friendliness, consumption of PV generation, etc. Users can also remotely control the operation of the equipment through the mobile terminal.

- Interaction with other households. The intelligent agent is allowed to interact with other households for the P2P power trading, thus to achieve the goal of on-site consumption of PV generation.
- Interaction with flexible loads. The intelligent agent communicates with flexible loads to obtain the monitored operation data of these domestic appliances and sent the control signal to automatically control the operation of the appliances.

Appendix B

The gray models for ACs are described by Eqs. (B.1)-(B.3). The model parameters include $\boldsymbol{b}_1, \boldsymbol{b}_2, \boldsymbol{b}_3, \boldsymbol{b}_4, \boldsymbol{b}_5, \boldsymbol{P}_{ac}^{on}, \boldsymbol{P}_{ac}^{off}$, and $\boldsymbol{\delta}_{ac}$. According to Dataset 1, the parameters $\boldsymbol{b}_1$, $\boldsymbol{b}_2$, $\boldsymbol{b}_3$, $\boldsymbol{b}_4$, and $\boldsymbol{b}_5$ were determined by linear regression methods. $\boldsymbol{P}_{ac}^{on}$, $\boldsymbol{P}_{ac}^{off}$, and $\boldsymbol{\delta}_{ac}$ are AC's operating power in ON state, AC's operating power in OFF state, and the thermostat dead-band, which were identified by statistical analysis based on Dataset 1.

$$\boldsymbol{T}_{air}(\boldsymbol{t}+1) = \boldsymbol{b}_1 \boldsymbol{T}_{air}(\boldsymbol{t}) + \boldsymbol{b}_2 \boldsymbol{T}_{out}(\boldsymbol{t}) \pm \boldsymbol{b}_3 \boldsymbol{P}_{ac}^{in}(\boldsymbol{t}) + \boldsymbol{b}_4 \boldsymbol{I}_s(\boldsymbol{t}) + \boldsymbol{b}_5 \tag{B.1}$$

$$\boldsymbol{P}_{ac}^{in}(\boldsymbol{t}) = \boldsymbol{U}_{ac}(\boldsymbol{t}) \cdot \boldsymbol{P}_{ac}^{on} + [1 - \boldsymbol{U}_{ac}(\boldsymbol{t})] \cdot \boldsymbol{P}_{ac}^{off} \tag{B.2}$$

$$\boldsymbol{U}_{ac}(\boldsymbol{t}) = \begin{cases} 0,\ \boldsymbol{T}_{air}(t) < \boldsymbol{T}_{ac}^{set}(t) - \dfrac{\boldsymbol{\delta}_{ac}}{2} \\ 1,\ \boldsymbol{T}_{air}(t) > \boldsymbol{T}_{ac}^{set}(t) + \dfrac{\boldsymbol{\delta}_{ac}}{2} \\ \boldsymbol{U}_{ac}(t-1), \boldsymbol{T}_{ac}^{set}(t) - \dfrac{\boldsymbol{\delta}_{ac}}{2} \le \boldsymbol{T}_{air}(t) \le \boldsymbol{T}_{ac}^{set}(t) + \dfrac{\boldsymbol{\delta}_{ac}}{2} \end{cases} \tag{B.3}$$

where $\boldsymbol{T}_{air}$ is indoor air temperature of an air-conditioning room, $\boldsymbol{T}_{out}$ is the outdoor ambient temperature, $\boldsymbol{P}_{ac}^{in}$ is operating power, $\boldsymbol{I}_s$ is the solar radiation, $\boldsymbol{U}_{ac}$ is a binary variable, $\boldsymbol{T}_{ac}^{set}$ is AC's temperature setpoint.

EWH's model are expressed by Eqs. (B.4)-(B.6). The model parameters are $\boldsymbol{c}_1$, $\boldsymbol{c}_2$, $\boldsymbol{c}_3$, $\boldsymbol{c}_4$, $\boldsymbol{c}_5$, $\boldsymbol{P}_{ewh}^{on}$, and $\boldsymbol{\delta}_{ewh}$, which can be identified by the same methods mentioned above and can also be adjusted adaptively with the updated Dataset 1.

$$\boldsymbol{T}_{w,in}(\boldsymbol{t}+1) = \boldsymbol{c}_1 \boldsymbol{T}_{w,in}(\boldsymbol{t}) + \boldsymbol{c}_2 \boldsymbol{T}_{en}(\boldsymbol{t}) + \boldsymbol{c}_3 \boldsymbol{P}_{ewh}^{in}(\boldsymbol{t}) + \boldsymbol{c}_4 \boldsymbol{P}_{use}(\boldsymbol{t}) + \boldsymbol{c}_5 \tag{B.4}$$

$$\boldsymbol{P}_{ewh}^{in}(\boldsymbol{t}) = \boldsymbol{U}_{ewh}(\boldsymbol{t}) \cdot \boldsymbol{P}_{ewh}^{on} \tag{B.5}$$

$$\boldsymbol{U}_{ewh}(\boldsymbol{t}) = \begin{cases} 0,\ \boldsymbol{T}_{w,in}(t) \ge \boldsymbol{T}_{ewh}^{set}(t) \\ 1, \boldsymbol{T}_{w,in}(\boldsymbol{t}) \le \boldsymbol{T}_{ewh}^{set}(t) - \boldsymbol{\delta}_{ewh} \\ \boldsymbol{U}_{ewh}(\boldsymbol{t}-1), \boldsymbol{T}_{ewh}^{set}(t) - \boldsymbol{\delta}_{ewh} < \boldsymbol{T}_{w,in}(\boldsymbol{t}) < \boldsymbol{T}_{ewh}^{set}(t) \end{cases} \tag{B.6}$$

where $\boldsymbol{T}_{w,in}$ is the hot water temperature, $\boldsymbol{T}_{en}$ is the surrounding environment temperature of EWHs, $\boldsymbol{P}_{ewh}^{in}$ denotes operating power, $\boldsymbol{P}_{use}$ represents the cooling derived from the supplementary cold water, $\boldsymbol{U}_{ewh}$ is a binary variable, $\boldsymbol{P}_{ewh}^{on}$ is EWH's heating power, $\boldsymbol{T}_{ewh}^{set}$ is temperature setpoint, $\boldsymbol{\delta}_{ewh}$ denotes thermostat dead-band.

The models of postponed loads, such as washing machines, clothes dryers, and dishwashers, were developed considering their real-life operation characteristics in our previous study [57], as formulated by Eqs. (B.7) and (B.8). A data-driven method was also proposed to identify the model parameters $P_{shift}^{i,m}$ and $\Delta t_{shift}^{i,m}$, particularly in practical engineering scenarios involving numerous postponed loads. The method utilizes a K-means clustering approach to identify the power profiles of postponed loads across various operation modes, based on history operation data. Subsequently, the operating power of different operation stages for each operation mode can be determined based on the clustered power profiles. Specifically, the average power of clustered power profiles at a particular operation stage and mode was employed as the postponed load's operating power for the corresponding stage and mode, as described in Eq. (B.9). Additionally, the duration of each stage $\Delta \tau_{shift}^{j,w}$ can be extracted from the clustered power curves. The model is designed to adaptively adjust these parameters, enhancing its accuracy with updated data from Dataset 1.

$$P_{shift}^{in}(t) = \begin{cases} P_{shift}^{i,1},\ t \in \left[t_{shift}^{start}, t_{shift}^{start} + \Delta t_{shift}^{i,1}\right] \\ P_{shift}^{i,2},\ t \in \left[t_{shift}^{start} + \Delta t_{shift}^{i,1}, t_{shift}^{start} + \Delta t_{shift}^{i,1} + \Delta t_{shift}^{i,2}\right] \\ \cdots\cdots \\ P_{shift}^{i,m},\ t \in \left[t_{shift}^{start} + \Delta t_{shift}^{i,1} + \ldots + \Delta t_{shift}^{i,m-1}, t_{shift}^{start} + \Delta t_{shift}^{i,1} + \ldots + \Delta t_{shift}^{i,m}\right] \\ 0,\ other \end{cases} \tag{B.7}$$

$$\Delta t_{shift}^{i} = \sum_{m=1}^{M} \Delta t_{shift}^{i,m} \tag{B.8}$$

$$P_{shift}^{i,m} = \frac{1}{H} \sum_{h=1}^{H} P_{shift}^{i,m,h} \tag{B.9}$$

where i is the operation mode of postponed loads; m is the operation stage of postponed loads; $P_{shift}^{i,m}$ is the operating power; $\Delta t_{shift}^{i,m}$ is the duration of operation stage m; t_{shift}^{start} is the start-up time; Δt_{shift}^{i} is the duration of the operation mode i; $P_{shift}^{i,m,h}$ is the h^{th} operating power sample of the clustered power profiles, and H is the total number of the sample at operation stage m of operation model i.

Appendix C

A comparison between the simulation results and the test results is presented in Fig. C.1.

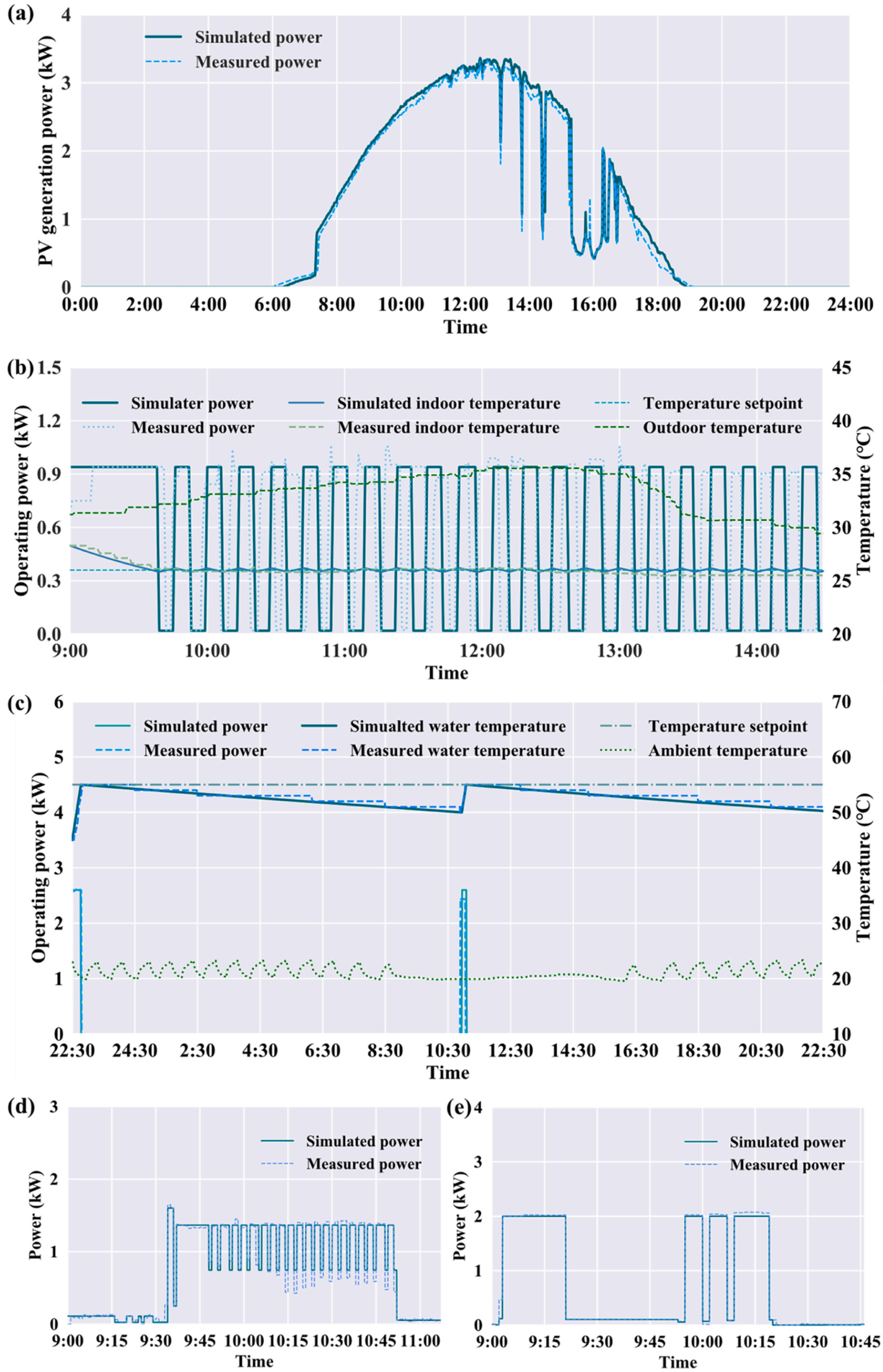

Fig. C.1. Validation of system models: (a) PV; (b) AC; (c) EWH; (d) washing and dryer machine; (e) dishwasher.

References

[1] The Paris Agreement | UNFCCC n.d. https://unfccc.int/process-and-meetings/the-paris-agreement/the-paris-agreement (accessed March 29, 2021).

[2] Langevin J, Harris CB, Satre-Meloy A, Chandra-Putra H, Speake A, Present E, et al. US building energy efficiency and flexibility as an electric grid resource. Joule 2021;5:2102–28. https://doi.org/10.1016/j.joule.2021.06.002.

[3] Wald D, Johnson K, King J, Comden J, Bay CJ, Chintala R, et al. Shifting demand: reduction in necessary storage capacity through tracking of renewable energy generation. Adv Appl Energy 2023;10:100131. https://doi.org/10.1016/j.adapen.2023.100131.

[4] Wu D, Zheng X, Menati A, Smith L, Xia B, Xu Y, et al. How much demand flexibility could have spared texas from the 2021 outage? Adv Appl Energy 2022;7. https://doi.org/10.1016/j.adapen.2022.100106.

[5] Luo Z., Peng J., Cao J., Yin R., Zou B., Tan Y., et al. Demand flexibility of residential buildings : definitions, flexible Loads, and quantification methods. Engineering 2022. https://doi.org/10.1016/j.eng.2022.01.010.

[6] Li H, Wang Z, Hong T, Piette MA. Energy flexibility of residential buildings: a systematic review of characterization and quantification methods and applications. Adv Appl Energy 2021;3:100054. https://doi.org/10.1016/j.adapen.2021.100054.

[7] Zhou B, Li W, Chan KW, Cao Y, Kuang Y, Liu X, et al. Smart home energy management systems: concept, configurations, and scheduling strategies. Renew Sustain Energy Rev 2016;61:30–40. https://doi.org/10.1016/j.rser.2016.03.047.

[8] Tuomela S, de Castro Tomé M, Iivari N, Svento R. Impacts of home energy management systems on electricity consumption. Appl Energy 2021;299. https://doi.org/10.1016/j.apenergy.2021.117310.

[9] Han B, Zahraoui Y, Mubin M, Mekhilef S, Seyedmahmoudian M, Stojcevski A. Home Energy Management Systems: a Review of the Concept, Architecture, and Scheduling Strategies. IEEE Access 2023;11:19999–20025. https://doi.org/10.1109/ACCESS.2023.3248502.

[10] Langer L, Volling T. An optimal home energy management system for modulating heat pumps and photovoltaic systems. Appl Energy 2020;278:115661. https://doi.org/10.1016/j.apenergy.2020.115661.

[11] Huy THB, Dinh HT, Kim D. Multi-objective framework for a home energy management system with the integration of solar energy and an electric vehicle using an augmented ε-constraint method and lexicographic optimization. Sustain Cities Soc 2023;88:104289. https://doi.org/10.1016/j.scs.2022.104289.

[12] Dinh HT, haeng Lee K, Kim D. Supervised-learning-based hour-ahead demand response for a behavior-based home energy management system approximating MILP optimization. Appl Energy 2022;321:119382. https://doi.org/10.1016/j.apenergy.2022.119382.

[13] Wang X, Mao X, Khodaei H. A multi-objective home energy management system based on internet of things and optimization algorithms. J Build Eng 2021;33:101603. https://doi.org/10.1016/j.jobe.2020.101603.

[14] Tostado-Véliz M, Kamel S, Aymen F, Jurado F. A novel hybrid lexicographic-IGDT methodology for robust multi-objective solution of home energy management systems. Energy 2022;253. https://doi.org/10.1016/j.energy.2022.124146.

[15] Tostado-Véliz M, Gurung S, Jurado F. Efficient solution of many-objective Home Energy Management systems. Int J Electr Power Energy Syst 2022;136. https://doi.org/10.1016/j.ijepes.2021.107666.

[16] Tostado-Véliz M, Hasanien HM, Kamel S, Turky RA, Jurado F, Elkadeem MR. Multiobjective home energy management systems in nearly-zero energy buildings under uncertainties considering vehicle-to-home: a novel lexicographic-based stochastic-information gap decision theory approach. Electr Power Syst Res 2023;214. https://doi.org/10.1016/j.epsr.2022.108946.

[17] Jin X, Baker K, Christensen D, Isley S. Foresee: a user-centric home energy management system for energy efficiency and demand response. Appl Energy 2017;205:1583–95. https://doi.org/10.1016/j.apenergy.2017.08.166.

[18] Tostado-Véliz M, León-Japa RS, Jurado F. Optimal electrification of off-grid smart homes considering flexible demand and vehicle-to-home capabilities. Appl Energy 2021;298. https://doi.org/10.1016/j.apenergy.2021.117184.

[19] Zheng Z, Sun Z, Pan J, Luo X. An integrated smart home energy management model based on a pyramid taxonomy for residential houses with photovoltaic-battery systems. Appl Energy 2021;298:117159. https://doi.org/10.1016/j.apenergy.2021.117159.

[20] Killian M, Zauner M, Kozek M. Comprehensive smart home energy management system using mixed-integer quadratic-programming. Appl Energy 2018;222:662–72. https://doi.org/10.1016/j.apenergy.2018.03.179.

[21] Duman AC, Erden HS, Gönül Ö, Güler Ö. A home energy management system with an integrated smart thermostat for demand response in smart grids. Sustain Cities Soc 2021;65. https://doi.org/10.1016/j.scs.2020.102639.

[22] Zupančič J, Filipič B, Gams M. Genetic-programming-based multi-objective optimization of strategies for home energy-management systems. Energy 2020;203. https://doi.org/10.1016/j.energy.2020.117769.

[23] Al Essa MJM. Home energy management of thermostatically controlled loads and photovoltaic-battery systems. Energy 2019;176:742–52. https://doi.org/10.1016/j.energy.2019.04.041.

[24] Duman AC, Erden HS, Gönül Ö, Güler Ö. Optimal sizing of PV-BESS units for home energy management system-equipped households considering day-ahead load scheduling for demand response and self-consumption. Energy Build 2022;267:112164. https://doi.org/10.1016/j.enbuild.2022.112164.

[25] Yousefi M, Hajizadeh A, Soltani MN, Hredzak B, Kianpoor N. Profit assessment of home energy management system for buildings with A-G energy labels. Appl Energy 2020;277:115618. https://doi.org/10.1016/j.apenergy.2020.115618.

[26] Ben Arab M, Rekik M, Krichen L. Suitable various-goal energy management system for smart home based on photovoltaic generator and electric vehicles. J Build Eng 2022;52:104430. https://doi.org/10.1016/j.jobe.2022.104430.

[27] Rocha HRO, Honorato IH, Fiorotti R, Celeste WC, Silvestre LJ, Silva JAL. An Artificial Intelligence based scheduling algorithm for demand-side energy management in Smart Homes. Appl Energy 2021;282. https://doi.org/10.1016/j.apenergy.2020.116145.

[28] Correa-Florez CA, Gerossier A, Michiorri A, Kariniotakis G. Stochastic operation of home energy management systems including battery cycling. Appl Energy 2018;225:1205–18. https://doi.org/10.1016/j.apenergy.2018.04.130.

[29] Alıç O, Filik ÜB. A multi-objective home energy management system for explicit cost-comfort analysis considering appliance category-based discomfort models and demand response programs. Energy Build 2021;240:110868. https://doi.org/10.1016/j.enbuild.2021.110868.

[30] Gomes ILR, Ruano MG, Ruano AE. MILP-based model predictive control for home energy management systems: a real case study in Algarve. Portugal. Energy Build 2023;281:112774. https://doi.org/10.1016/j.enbuild.2023.112774.

[31] Gonçalves I, Á Gomes, Henggeler Antunes C. Optimizing the management of smart home energy resources under different power cost scenarios. Appl Energy 2019;242:351–63. https://doi.org/10.1016/j.apenergy.2019.03.108.

[32] Esmaeel Nezhad A, Rahimnejad A, Gadsden SA. Home energy management system for smart buildings with inverter-based air conditioning system. Int J Electr Power Energy Syst 2021;133:107230. https://doi.org/10.1016/j.ijepes.2021.107230.

[33] Wu H, Pratt A, Munankarmi P, Lunacek M, Balamurugan SP, Liu X, et al. Impact of model predictive control-enabled home energy management on large-scale distribution systems with photovoltaics. Adv Appl Energy 2022;6:100094. https://doi.org/10.1016/j.adapen.2022.100094.

[34] Ajao A., Luo J., Liang Z., Alsafasfeh Q.H., Su W. Intelligent home energy management system for distributed renewable generators, dispatchable residential loads and distributed energy storage devices. 2017 8th Int Renew Energy Congr IREC 2017 2017:1–6. https://doi.org/10.1109/IREC.2017.7926040.

[35] Amabile L, Bresch-Pietri D, El Hajje G, Labbé S, Petit N. Optimizing the self-consumption of residential photovoltaic energy and quantification of the impact of production forecast uncertainties. Adv Appl Energy 2021;2:100020. https://doi.org/10.1016/j.adapen.2021.100020.

[36] Javadi MS, Gough M, Lotfi M, Esmaeel Nezhad A, Santos SF, Catalão JPS. Optimal self-scheduling of home energy management system in the presence of photovoltaic power generation and batteries. Energy 2020;210:118568. https://doi.org/10.1016/j.energy.2020.118568.

[37] Javadi MS, Nezhad AE, Nardelli PHJ, Gough M, Lotfi M, Santos S, et al. Self-scheduling model for home energy management systems considering the end-users discomfort index within price-based demand response programs. Sustain Cities Soc 2021;68. https://doi.org/10.1016/j.scs.2021.102792.

[38] Nezhad A.E., Nardelli P.H.J., Sahoo S., Ghanavati F., Osorio G.J. A centralized home energy management system to minimize consumer's electricity bill. 2022 IEEE Int Conf Environ Electr Eng 2022 IEEE Ind Commer Power Syst Eur EEEIC /I CPS Eur 2022 2022. https://doi.org/10.1109/EEEIC/ICPSEurope54979.2022.9854599.

[39] Sahoo S, Ghanavati FA. Shrinking Horizon Model Predictive Controller for Daily Scheduling of Home Energy Management Systems. IEEE Access 2022;10:29716–30.

[40] Elkazaz M, Sumner M, Naghiyev E, Pholboon S, Davies R, Thomas D. A hierarchical two-stage energy management for a home microgrid using model predictive and real-time controllers. Appl Energy 2020;269:115118. https://doi.org/10.1016/j.apenergy.2020.115118.

[41] Liu Y, Ma J, Xing X, Liu X, Wang W. A home energy management system incorporating data-driven uncertainty-aware user preference. Appl Energy 2022;326:119911. https://doi.org/10.1016/j.apenergy.2022.119911.

[42] Mani M, Pillai R. Impact of dust on solar photovoltaic (PV) performance: research status, challenges and recommendations. Renew Sustain Energy Rev 2010;14:3124–31. https://doi.org/10.1016/j.rser.2010.07.065.

[43] Salameh W, Faraj J, Harika E, Murr R, Khaled M. On the optimization of electrical water heaters: modelling simulations and experimentation. Energies 2021;14:1–12. https://doi.org/10.3390/en14133912.

[44] Sabry AH, Shallal AH, Hameed HS, Ker PJ. Compatibility of household appliances with DC microgrid for PV systems. Heliyon 2020;6. https://doi.org/10.1016/j.heliyon.2020.e05699.

[45] Duan J, Li N, Peng J, Wang C, Liu Q, Zhou X. Study on occupant behaviour using air conditioning of high-rise residential buildings in hot summer and cold winter zone in China. Energy Build 2022;276:112498. https://doi.org/10.1016/j.enbuild.2022.112498.

[46] Vivian J, Chiodarelli U, Emmi G, Zarrella A. A sensitivity analysis on the heating and cooling energy flexibility of residential buildings. Sustain Cities Soc 2020;52:101815. https://doi.org/10.1016/j.scs.2019.101815.

[47] Bünning F, Heer P, Smith RS, Lygeros J. Increasing electrical reserve provision in districts by exploiting energy flexibility of buildings with robust model predictive control. Adv Appl Energy 2023;10. https://doi.org/10.1016/j.adapen.2023.100130.

[48] Zhang K, Prakash A, Paul L, Blum D, Alstone P, Zoellick J, et al. Model predictive control for demand flexibility: real-world operation of a commercial building with photovoltaic and battery systems. Adv Appl Energy 2022;7:100099. https://doi.org/10.1016/j.adapen.2022.100099.

[49] Gu W, Liu S, Zhang Z, Li Y. A distributed physical architecture and data-based scheduling method for smart factory based on intelligent agents. J Manuf Syst 2022;65:785–801. https://doi.org/10.1016/j.jmsy.2022.11.006.

[50] Li P, Anduv B, Zhu X, Jin X, Du Z. Across working conditions fault diagnosis for chillers based on IoT intelligent agent with deep learning model. Energy Build 2022;268:112188. https://doi.org/10.1016/j.enbuild.2022.112188.
[51] Peng J, Luo Z, Tan Y, Jiang H, Yin R, Yan J. Balancing stakeholder benefits: a many-objective optimal dispatch framework for home energy systems inspired by Maslow's Hierarchy of needs. Adv Appl Energy 2024;13. https://doi.org/10.1016/j.adapen.2023.100160.
[52] Wooldridge, Michael and NRJ. Intelligent Agents: theory and Practice 1995.
[53] Luo Z, Peng J, Yin R. Many-objective day-ahead optimal scheduling of residential flexible loads integrated with stochastic occupant behavior models. Appl Energy 2023;347:121348. https://doi.org/10.1016/j.apenergy.2023.121348.
[54] Wang Z, Sun M, Gao C, Wang X, Chris B. A new interactive real-time pricing mechanism of demand response based on an evaluation model. Appl Energy 2021; 295:117052. https://doi.org/10.1016/j.apenergy.2021.117052.
[55] Deb K, Jain H. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, Part I: solving problems with box constraints. IEEE Trans Evol Comput 2014;18:577–601. https://doi.org/10.1109/TEVC.2013.2281535.
[56] Luo Z, Yang S, Xie N, Xie W, Liu J, Souley Agbodjan Y, et al. Multi-objective capacity optimization of a distributed energy system considering economy, environment and energy. Energy Convers Manag 2019;200:112081. https://doi.org/10.1016/j.enconman.2019.112081.
[57] Luo Z, Peng J, Hu M, Liao W, Fang Y. Multi-objective optimal dispatch of household flexible loads based on their real-life operating characteristics and energy-related occupant behavior. Build Simul 2023.